1 Basics of Signals

Objectives

Upon completion of this chapter, you will be able to:

1. Understand various classifications of signals.
2. Understand and apply different operations on dependent and independent variables.
3. Understand basic elementary signals, standard signals, doublet function, and its properties.
4. Obtain the graphical representation of the signal for the given x(t)/x(n).
5. Write equivalent time domain expression in terms of elementary signals from the graph of x(t)/x(n).
6. Convert sinc function to sampling function and vice-versa, understand its properties.
7. Try questions to test their learning.
8. Make a note of the important points and mistakes usually committed.

Introduction

This is the chapter of utmost importance. In this chapter, we will be learning about signals, their physical and mathematical representations in detail, doing synthesis, and obtaining the signal using basic mathematical concepts, and doing analysis of both continuous-time signals and discrete-time signals by introducing mathematical descriptions and equations, which will indeed be carried forward in the upcoming chapters considerably.

Signal: Signal is a mathematical representation of a physical phenomenon, which contains some information.

Example:

Physical phenomenon: AC Voltage

Mathematical rep: $x(t) = A\cos\omega_0 t$

Signal is a dependent variable which depends on one or more independent variables.

Signal	Dependent Variable	Independent Variable	Name
x(t)	x(t)	t → Continuous-time	Continuous-time Domain signal
x[n]	x[n]	n → Discrete-time	Discrete-time domain signal
$X(\omega)$ or X(f)	$X(\omega)$ or X(f)	ω or f → Continuous frequency	Continuous frequency domain signal
$X(n\omega_0)$ or $X(nf_0)$	$X(n\omega_0)$ or $X(nf_0)$	$n\omega_0$ or nf_0 → Discrete frequency	Discrete frequency domain signal

Table 1.1

Classification of signals:

1. Continuous-time and discrete-time signals
2. Even and odd signals
3. Even conjugate and odd conjugate signals
4. Periodic and non-periodic signals
5. Orthogonal and non-orthogonal signals
6. Energy and power signals
7. Causal and non-causal signals
8. Bounded and unbounded signals
9. Absolutely integrable/summable signals
10. Left-sided, right-sided, and both-sided signals (LSS, RSS, and BSS).

Continuous-Time and Discrete-Time Signals

Continuous-time signal (CTS):

1. Continuous in amplitude: [LHL = RHL = f(a)] → Finite

2. Continuous in time: Continuous time signals have independent variable t, and are represented as x(t) vs t.

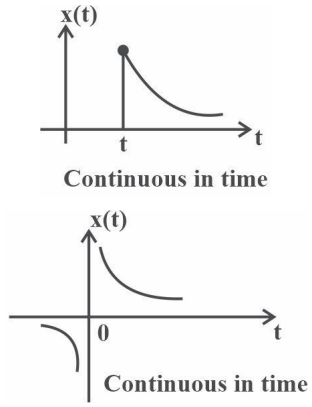


Fig. 1.1

Discrete-time signal (DTS):

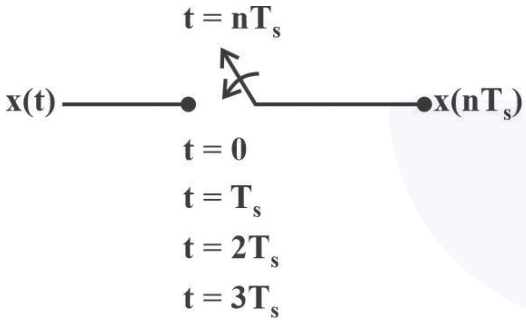


Fig. 1.2

Discrete-time signals are obtained from continuous-time signals by a process called "Uniform Sampling".

Where,

T_s = Sampling interval

$f_s = 1/T_s$ = Sampling frequency or sampling rate

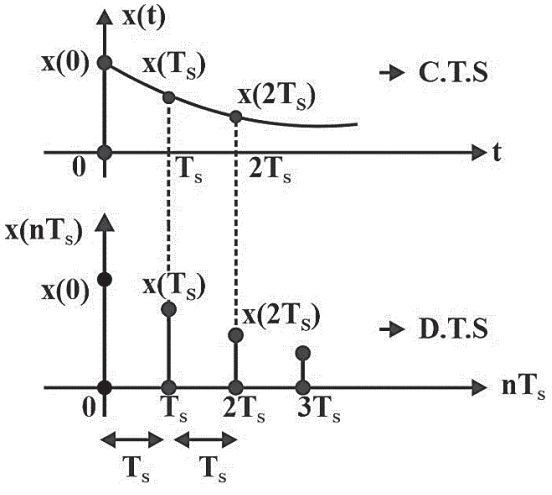


Fig. 1.3

Mathematical conversion from CTS to DTS:

Given x(t) = f(t): $a \le t \le b$

Step 1: Identify f_s or T_s

Step 2: Put $t = nT_s$

$x(nT_s) = f(nT_s): a \le nT_s \le b$

$x(n) = f(nT_s): \frac{a}{T_s} \le n \le \frac{b}{T_s}$

Example 1.1: x(t) = sin (200πt) , calculate discrete-time signal if sampling rate is 10Hz.

Solution:

Step 1: $T_s = \frac{1}{f_s} = 0.1\text{sec}$

Step 2: $x(nT_s) = \sin\left(200\pi\left(nT_s\right)\right)$

$x(n) = \sin(20\pi n)$

Operations on Signals

Operations on continuous-time signal:

Operations on DV:

Amplitude scaling:

Given: x(t) vs t

Plot: Ax(t) vs t

Process: Multiply vertical axis parameter by A.

Example 1.2: For the following x(t)Plot y(t) = 2x(t)

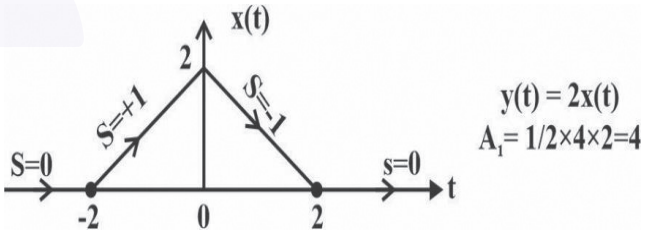


Fig. 1.4

Solution:

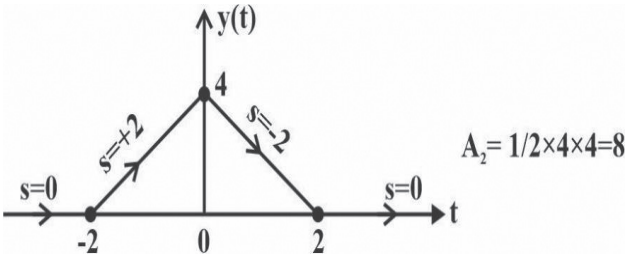


Fig. 1.5

Amplitude reversal:

Given: x(t) vs t

Plot: x(t) vs t

Procedure: Take mirror image w.r.t horizontal axis.

Example 1.3: For following x(t)Plot y(t) = -x(t)

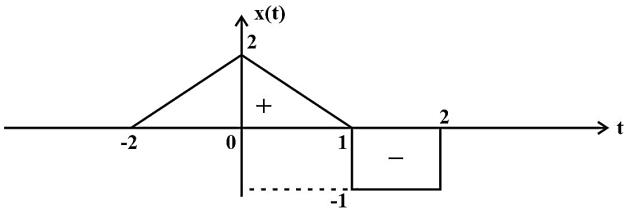


Fig. 1.6

Solution:

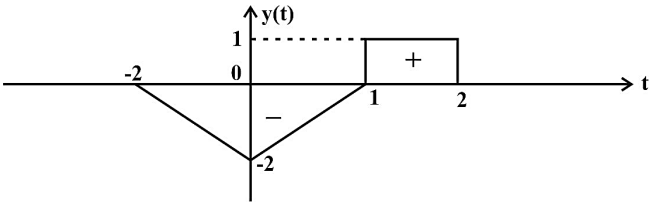


Fig. 1.7

Modulus operation:

Given: x(t) vs t

Plot: |x(t)| vs t

Procedure:

a) Above horizontal axis: Keep as it is

b) Below horizontal axis: Reflect above wrt horizontal axis.

Example 1.4: For following x(t)Plot y(t) = |x(t)|

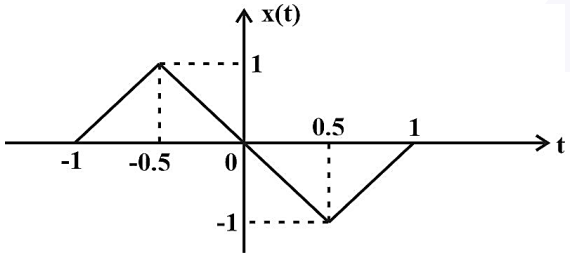


Fig. 1.8

Solution:

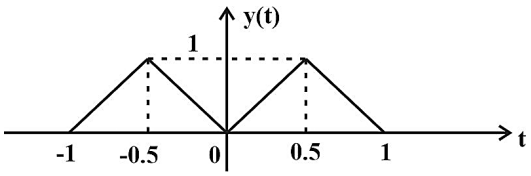


Fig. 1.9

Addition or subtraction of DC value:

Given: x(t) vs t

Plot: x(t) ± A vs t

Procedure:

a) X(t) + A → Shift A units upwards

b) X(t) - A → Shift A units downwards

Example 1.5: For the following x(t)Plot y(t) = 2 + x(t) and y(t) = x(t) - 2

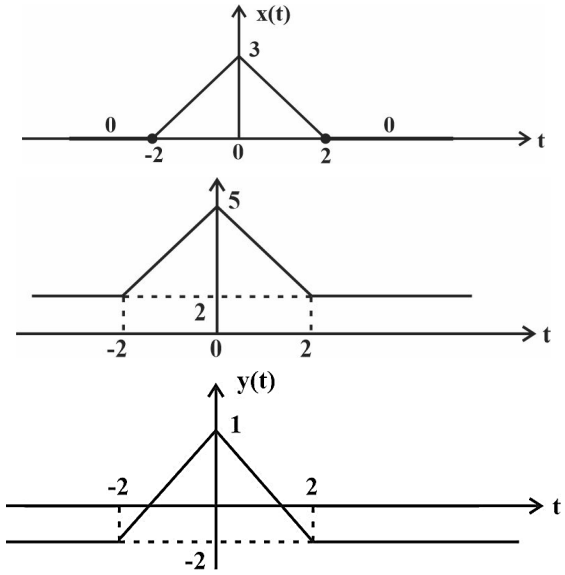


Fig. 1.10

Operations on Independent Variable

Time shifting:

Given: x(t) vs t

Plot:

a) $X(t-t_0)$ vs t → Right shift by t_0 → Delay operation

b) $X(t+t_0)$ vs t → Left shift by t_0 → Advance operation

Time scaling:

Given: x(t) vs t [a > 0]

Plot: x(at) vs t

Procedure: Divide horizontal axis by a

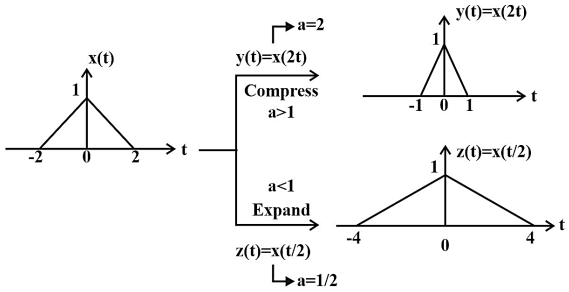


Fig. 1.11

Time reversal:

Given: x(t) vs t

Plot: x(-t) vs t

Procedure: Mirror image w.r.t vertical axis.

Example 1.6: For following x(t)Plot y(t) = x(-t)

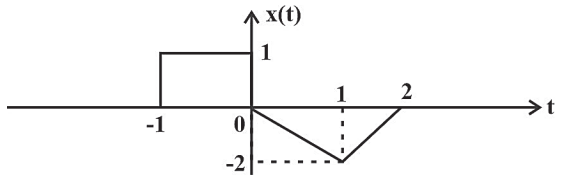


Fig. 1.12

Solution:

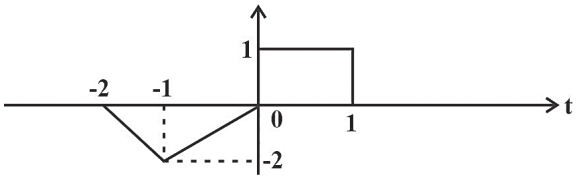


Fig. 1.13

Mixed operation:

Given: x(t) vs t

Plot: x(b-at) vs t Plot x(-3t+2)

Step a) Write in the format x(- at + b)

Step b)

a) $x(t) \xrightarrow{\text{Shifting}} x(t+2) \xrightarrow{\text{Scaling}}$

$x(3t+2) \rightarrow x(-3t+2)$

b) $x(t) \xrightarrow{\text{Scaling}} x(3t) \xrightarrow[t \rightarrow t+\frac{2}{3}]{\text{Shifting}}$

$x(3(t+\frac{2}{3})) = x(3t+2)$

↓ Reversal

x(-3t+2)

c) $x(t) \xrightarrow{\text{Reversal}} x(-t) \xrightarrow[t \rightarrow t-2]{\text{Shifting}}$

$x(-(t-2)) = x(-t+2)$

↓ Scaling

x(-3t+2)

Natural order: Shifting → Scaling → Reversal

Note: In CTS, any order of mixed operation can be followed. Preference is given to natural order.

Note: Precedence rule, shifting first scaling next. Other orders should not be considered.

Example 1.7:

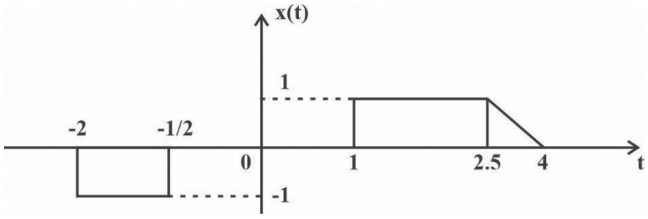


Fig. 1.14

$y(t) = x\left(\frac{1-2t}{3}\right) = ?$

Solution:

Step 1 → $x\left(-\frac{2t}{3}+\frac{1}{3}\right)$

$x(t) \longrightarrow x\left(t+\frac{1}{3}\right) \xrightarrow{a=2/3} x\left(\frac{2}{3}t+\frac{1}{3}\right)$

$\longrightarrow x\left(\frac{-2t}{3}+\frac{1}{3}\right)$

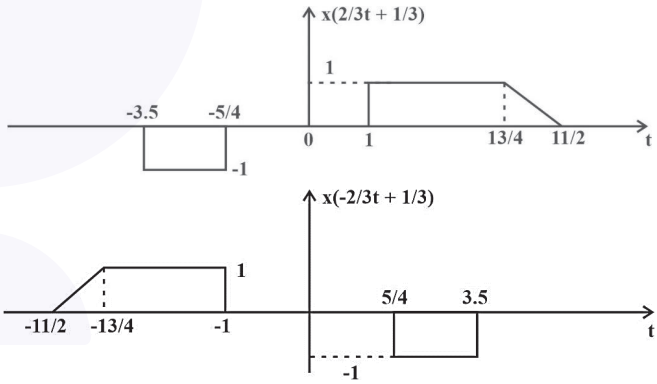


Fig. 1.15

Given: x(t) vs t

Plot: x(|t|) vs t

Procedure:

a) Left half side of graph is removed

b) Take mirror image of right-hand side w.r.t. vertical axis.

Addition and Subtraction of Signals

1. $x_1(t) + x_2(t)$
2. $x_1(t) - x_2(t) = x_1(t) + (-x_2(t))$

Rules of addition:

1. Track graph from -∞ to +∞ (Left to right).
2. Indicate points where definition of signal changes.
3. Indicate regions with slopes.
4. While tracking graph always holds previous position.
5. **Case 1:** At a point

$$\left.\begin{array}{l} x_1(t) \to p \uparrow \\ x_2(t) \to q \downarrow \end{array}\right|$$

→ net vertical effect is considered

Case 2: In an interval

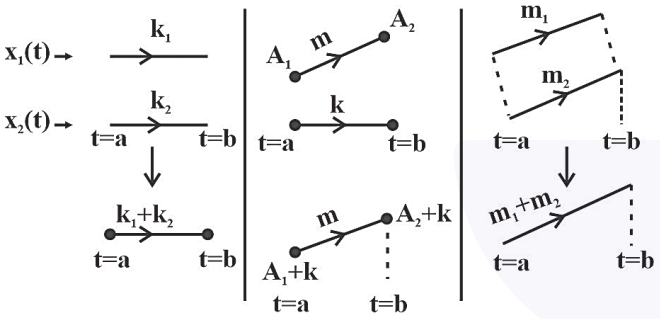


Fig. 1.16

Example 1.8:

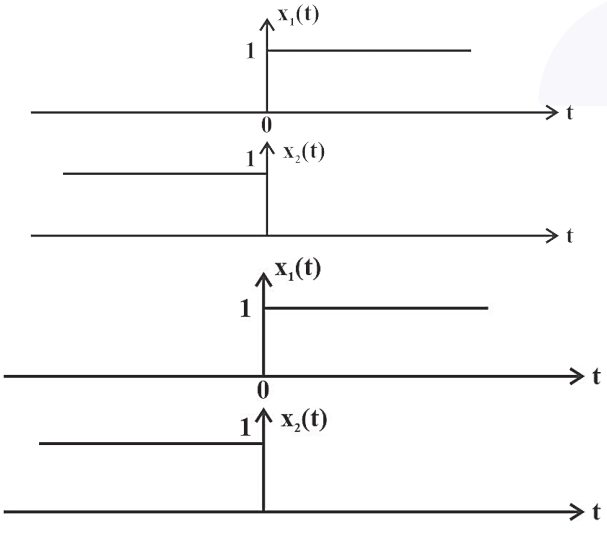


Fig. 1.17

Plot $x_1(t) + x_2(t) = ?$

Solution:

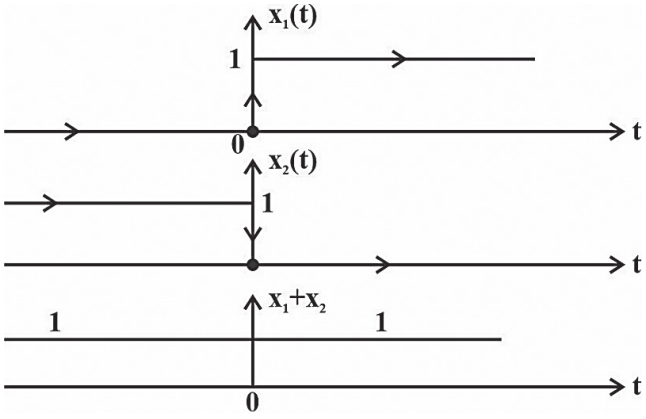


Fig. 1.18

Standard CTS

Argument:

x(-2t + 3) → Argument = -2t + 3

Breakpoint (BP):

Point at which argument = 0

t = 3/2 → B.P.

Approximation of a signal: After approximation of signal, area of signal does not change.

Area of $x(t) \to \int_{-\infty}^{\infty} x(t)dt = A$

Let x(t) is approximated as f(t)

$$\lim_{t \to a} x(t) = f(t)$$

$$\int_{-\infty}^{\infty} f(t)dt = A$$

Area of f(t) is approximated as area of x(t)

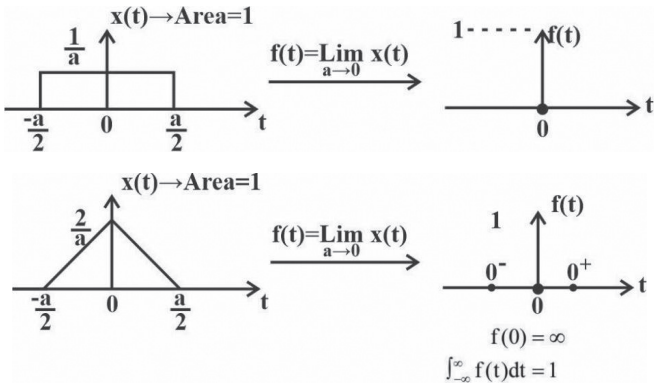


Fig. 1.19

Unit impulse signal:

Graph:

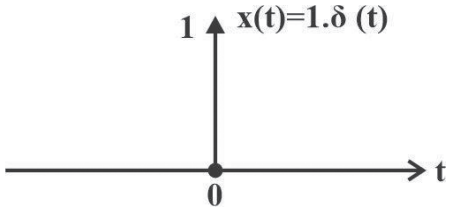


Fig. 1.20

Mathematical definition:

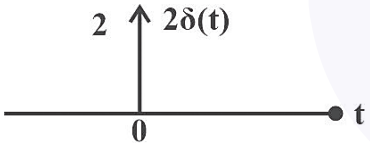
$$x(t) = \delta(t) = 1 \cdot \delta(t)$$

$$x(t) = \delta(t) = \begin{cases} 0 & t \neq 0 \\ \neq 0 & t = 0 \end{cases}$$

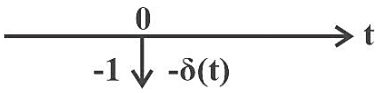
$$\int_{-\infty}^{\infty} \delta(t)dt = 1 \rightarrow \int_{0^-}^{0^+} \delta(t)dt = 1$$

Basic operation:

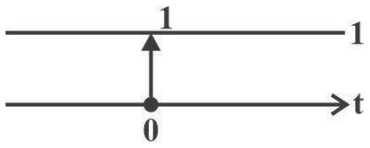
1. $2\delta(t)$



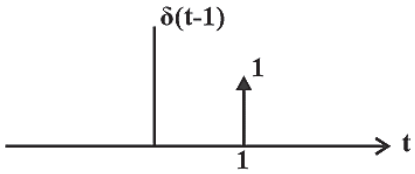
2. $-\delta(t)$



3. $\delta(t) + 1$



4. $\delta(t-1)$



5. $\delta(2t) = \frac{1}{2}\delta(t)$

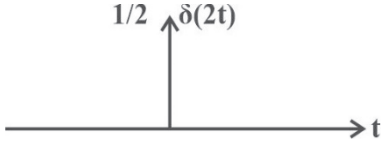


Fig. 1.21

Proof:

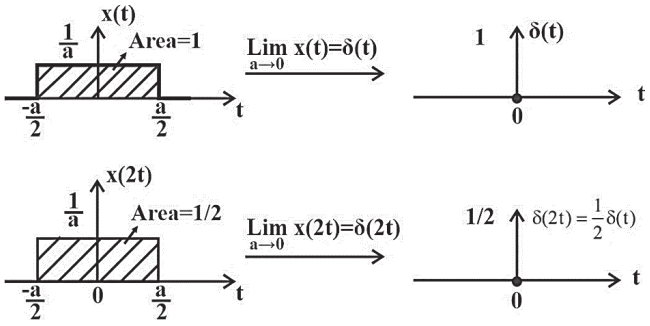


Fig. 1.22

6. $\delta(t) \rightarrow \delta(t+2) \rightarrow \delta(3t+2) \rightarrow \delta(-3t+2)$

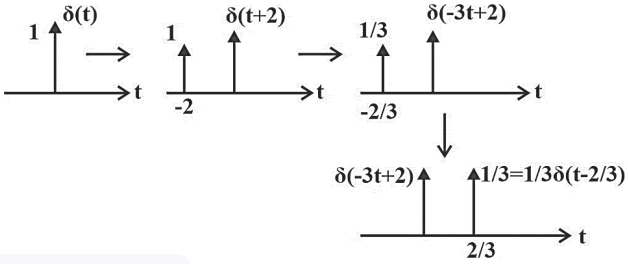


Fig. 1.23

$$\delta(-3t+2) = \frac{1}{3}\delta\left(t - \frac{2}{3}\right)$$

Properties of impulse signal:

1. $\delta(-t) = \delta(t) \rightarrow$ Impulse at origin is even signal
2. $\delta(t \pm t_0) \rightarrow$ Not even signal
3. $\delta(bt) = \frac{1}{|b|}\delta(t)$
4. $\delta(-bt) = \delta(bt) = \frac{1}{|b|}\delta(t)$
5. $\delta(at+b) = \delta\left(a\left(t + \frac{b}{a}\right)\right) = \frac{1}{|a|}\delta\left(t + \frac{b}{a}\right)$
6. $\delta(-at+b) = \delta\left(-a\left(t - \frac{b}{a}\right)\right) = \frac{1}{|-a|}\delta\left(t - \frac{b}{a}\right)$
7. $\delta(at-b) = \delta\left(a\left(t - \frac{b}{a}\right)\right) = \frac{1}{|a|}\delta\left(t - \frac{b}{a}\right)$
8. $\delta(-at-b) = \delta\left(-a\left(t + \frac{b}{a}\right)\right) = \frac{1}{|-a|}\delta\left(t + \frac{b}{a}\right)$

Note: Care should be taken while applying time scaling property. In the denominator, |a| will come where 'a' is the scaling factor.

Example 1.9: Find the area and breakpoint BP of the following signals.

a) $\delta(-t-1)$

Solution:

$$\delta(-t-1)=\delta\left(-(t+1)\right)=\frac{1}{|-1|}\delta(t+1)=\delta(t+1)$$

$BP: t=-1, \quad Area=1$

b) $\delta\left(\frac{1-2t}{3}\right)$

Solution:

$$\delta\left(\frac{1-2t}{3}\right)=\delta\left(\frac{-2t}{3}+\frac{1}{3}\right)=\delta\left(-\frac{2}{3}\left(t-\frac{1}{2}\right)\right)$$

$$=\frac{1}{\left|\frac{-2}{3}\right|}\delta\left(t-\frac{1}{2}\right)=\frac{3}{2}\delta\left(t-\frac{1}{2}\right)$$

$BP: t=\frac{1}{2}$

$Area=\frac{3}{2}$

9. $\delta[g(t)]=\sum_i \frac{\delta(t-t_i)}{|g'(t_i)|}\rightarrow$ where t_i is roots of g(t) = 0.

10. Sampling or multiplication property

$x(t)\delta(t)=x(0)\delta(t)$

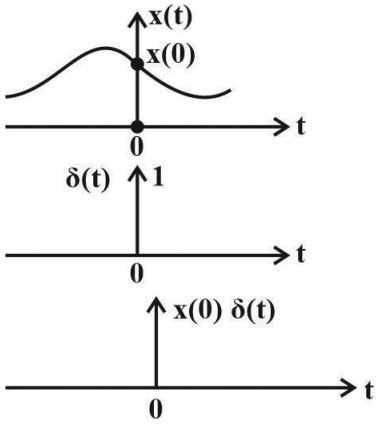


Fig. 1.24

Example 1.10: $x(t)\delta(2t)=?$

Solution: $x(t)\delta(2t)=x(t)\cdot\frac{1}{2}\delta(t)=\frac{1}{2}x(0)\delta(t)$

11. Integration property

$$I=\int_{t_1}^{t_2} x(t)\ \delta(-at+b)dt$$

$$I=\frac{1}{|-a|}\int_{t_1}^{t_2} x(t)\ \delta\left(t-\frac{b}{a}\right)dt\rightarrow$$

$$I=\frac{1}{a}\int_{t_1}^{t^2} x\left(\frac{b}{a}\right)\delta\left(t-\frac{b}{a}\right)dt$$

$$I=\frac{1}{a}x\left(\frac{b}{a}\right)\int_{t_1}^{t_2}\delta\left(t-\frac{b}{a}\right)dt$$

= 0 if impulse is not between $t_1 \le t \le t_2$

= 1 if impulse is present between $t_1 \le t \le t_2$

Example 1.11: $I=\int_{-\infty}^{\infty} x(t)\delta(t)dt=?$

Solution: $I=\int_{-\infty}^{\infty} x(t)\delta(t)dt$

$$I=\int_{-\infty}^{\infty} x(0)\delta(t)dt$$

$$I=x(0)\int_{-\infty}^{\infty}\delta(t)dt=x(0)\cdot 1$$

Example 1.12: $I=\int_{-4}^{6} x(t)\delta(-2t+1)dt\ =?$

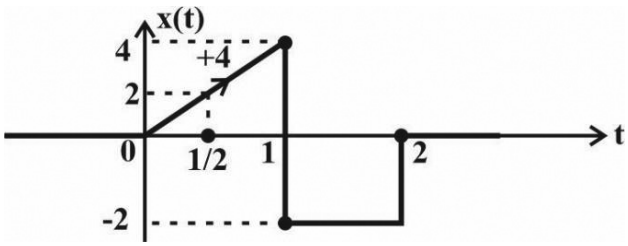


Fig. 1.25

Solution:

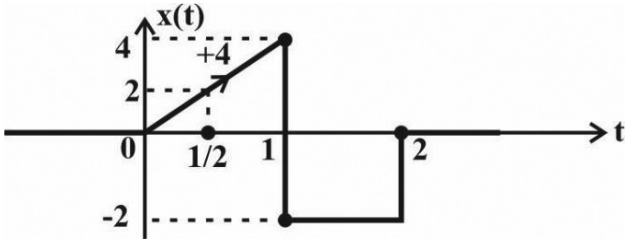


Fig. 1.26

$$I = \int_{-4}^{6} x(t)\delta(-2t+1)dt$$

$$= \frac{1}{2}\int_{-4}^{6} x(t)\delta\left(t-\frac{1}{2}\right)dt$$

$$\frac{1}{2}x\left(\frac{1}{2}\right)\int_{-4}^{6}\delta\left(t-\frac{1}{2}\right)dt = \frac{1}{2}x\left(\frac{1}{2}\right) = \frac{1}{2}\times 2 = 1$$

Note:

1. If the breakpoint of impulse signal is present within the limits of integration, then the integration is not defined.
2. If the breakpoint of the impulse signal is present within (between, i.e., excluding the limits itself) the limits of integration, then the integration will be defined.
3. If the breakpoint of the impulse signal is not present within the limit and in the limits of integration, then the value of the integration is zero.

Unit step signal:

Let us perform running integration of x(t)

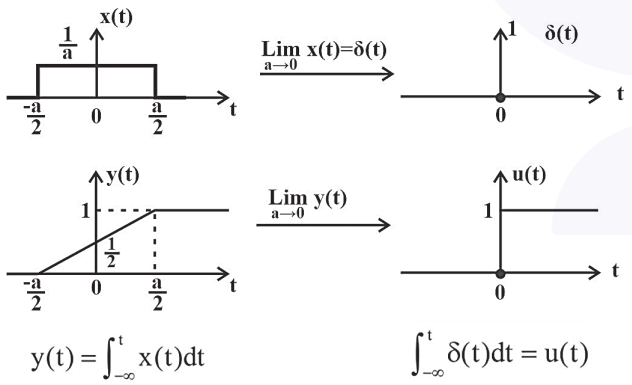


Fig. 1.27

Let us perform running integration of x(t)

$\int_{-\infty}^{\infty} x(t)dt \rightarrow$ Integration of x(t)

$\int_{-\infty}^{\infty} |x(t)|dt \rightarrow$ Absolute integration or absolute area

$\int_{-\infty}^{t} x(t)dt \rightarrow$ Running integration

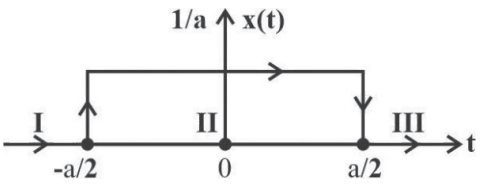


Fig. 1.28

$$y(t) = \int_{-\infty}^{t} x(t)dt$$

$$-\infty < t < -\frac{a}{2} \quad : \quad y(t) = \int_{-\infty}^{t} x(t)dt = 0$$

$$\frac{-a}{2} < t < \frac{a}{2} \quad : \quad y(t)$$

$$= \int_{-\infty}^{t} x(t)dt = \int_{-\infty}^{-a/2} x(t)dt$$

$$+\int_{-a/2}^{t} x(t)dt = \left(\frac{t}{a}+\frac{1}{2}\right)$$

$$= \int_{-\infty}^{t} x(t)dt = \int_{-\infty}^{-a/2} x(t)dt$$

$$+\int_{-a/2}^{a/2} x(t)dt + \int_{a/2}^{t} x(t)dt = 1$$

$$y(t) = \int_{-\infty}^{t} x(t)dt = \begin{cases} 0 & : \ -\infty < t < -a/2 \\ \frac{t}{a}+\frac{1}{2} & : \ -\frac{a}{2} < t < \frac{a}{2} \\ 1 & : \ \frac{a}{2} < t < \infty \end{cases}$$

Unit step signal:

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$

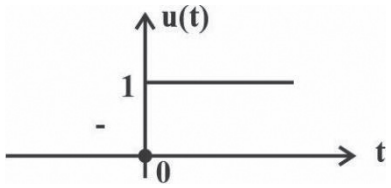


Fig. 1.29

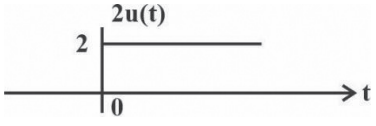
At t = 0:

1. Finite + uncertain = Indeterminate
2. Gibb's phenomenon: Average of preceding and succeeding $\frac{0+1}{2} = \frac{1}{2}$

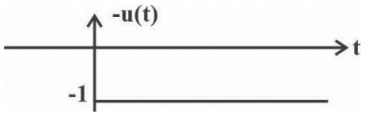
3. Working value: $u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$

Operation on step signal:

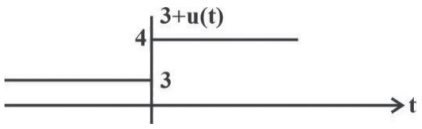
1. 2u(t)



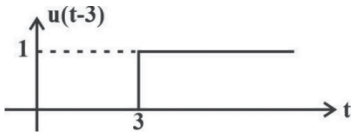
2. –u(t)



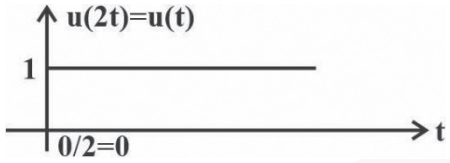
3. 3 + u(t)



4. u(t-3)



5. u(2t)



6. u(-2t + 3)

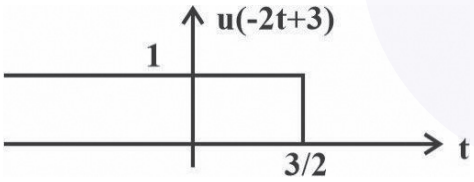
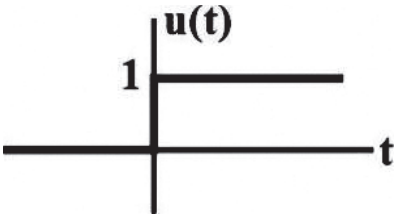


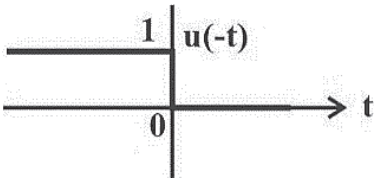
Fig. 1.30

Important observation:

1. $u(t)+u(-t)=1$



2. a) $u(t)-u(-t)=2u(t)-1$



b) $u(t)-u(-t)=\text{sgn}(t)=\begin{cases}1: t>0\\0: t=0\\-1: t<0\end{cases}$

$$=\begin{cases}\dfrac{|t|}{t}=\dfrac{t}{|t|}: t\neq 0\\ 0 \quad : t=0\end{cases}$$

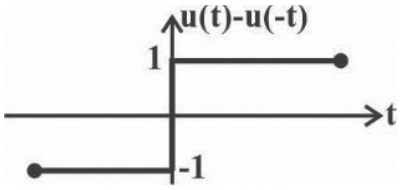


Fig. 1.31

3. u(at) = u(t)
4. u(t) u(-t) = 0
5. $u(t)\times u(t)\times\cdots\infty\text{times}=u(t)$
6. $u(-t)u(-t)\times\cdots\infty times=u(-t)$
7. $u(\sin t)=\begin{cases}1 & : \sin t\geq 0\\ 0 & : \sin t<0\end{cases}$

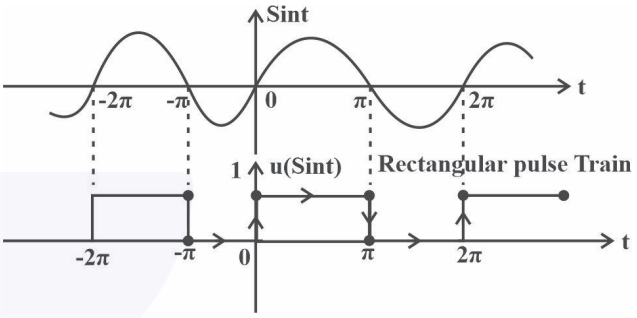


Fig. 1.32

Synthesis and Analysis

Synthesis:

Case: 1 When all the sign of t is same

Step a): Identify the BPStep

Step b): Indicate BP on horizontal axis

Step c): Add/Sub respective step sizes at BP

Example 1.13:

Plot $x(t)=1\cdot u(t)+1\cdot u(t-1)+1\cdot u(t-2)$

Solution: $x(t)=1\cdot u(t)+1\cdot u(t-1)+1\cdot u(t-2)$

B.P. are at t = 0, 1, and 2

Sign of t is positive, hence track → Left to right

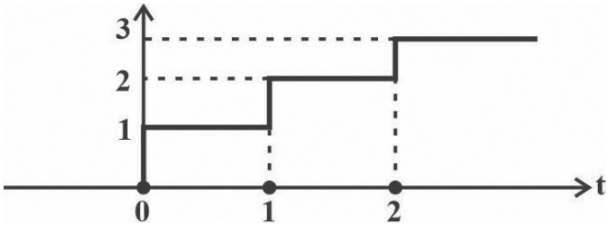


Fig. 1.33

Sum of amplitude coefficient $\neq 0 \rightarrow$ Finite amplitude + Infinite duration

Example 1.14:

Plot $x(t) = u(t) + u(t-1) + u(t-2) - 3u(t-3)$

Solution:

$$x(t) = u(t) + u(t-1) + u(t-2) - 3u(t-3)$$

B.P. are at t = 0, 1, 2, and 3

Sign of t is positive, hence track → Left to right

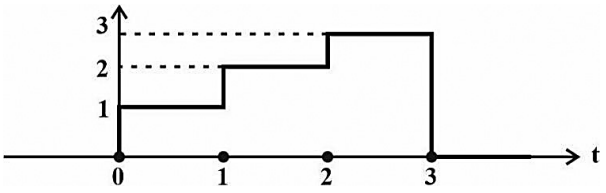


Fig. 1.34

Sum of amplitude coefficient $= 0 \rightarrow$ Finite duration + Finite amplitude

Conclusion:

1. If all the coefficient of t is positive track left to right.
2. If all the coefficient of t is negative track right to left.
3. When all the signs of t is same and sum of amplitude coefficient = 0.

   Then signal is → Finite duration + Finite amplitude

Case 2: When t is having different signs

a) Positive sign of t signal is plotted together

b) Negative sign of t signal is plotted together Then addition of two signals is performed

Example 1.15: Plot $x(t) = u(-2t+3) + u(-t-1) + u(t-1) + u(t-3/2) = ?$

Solution: $x(t) = u(-2t+3) + u(-t-1) + u(t-1) + u(t-3/2)$

BP are at $t = -1, 1, \frac{3}{2}$

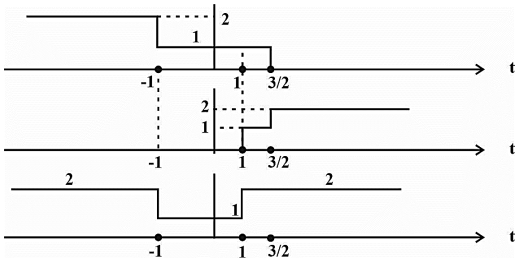


Fig. 1.35

Analysis:

If tracking is left to right:

Then at each B.P. signal = Step change at B.P. [u(t-B.P.)]

If tracking is right to left:

Then at each B.P. signal = Step change at B.P. [u(-t+B.P.)]

Example 1.16: Write the expression for the signal x(t) shown in figure below.

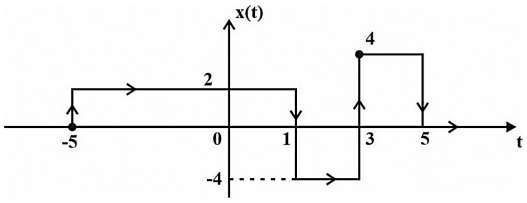


Fig. 1.36

Solution:

$$x(t) = 2u(t+5) - 6u(t-1) + 8u(t-3) - 4u(t-5)$$

Or

$$x(t) = 4u(-t+5) - 8u(-t+3) + 6u(-t+1) - 2u(-t-5)$$

Note: Proper subtraction of (final height-initial height) at that breakpoint is to be done.

Ramp signal:

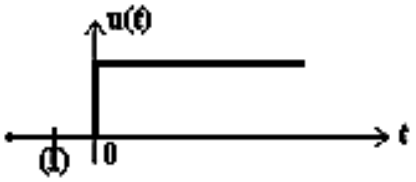
$$r(t) = \int_{-\infty}^{t} u(t)dt = \begin{cases} 0 & : \quad t < 0 \\ t & : \quad t \geq 0 \end{cases}$$

$$t < 0 : \int_{-\infty}^{t} u(t)dt = 0$$

$$t > 0 : \int_{-\infty}^{t} u(t)dt = \int_{-\infty}^{0} 0 \cdot dt + \int_{0}^{t} 1dt = t$$

$$\rightarrow r(t) = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases}$$

$$\rightarrow r(t) = t\ u(t)$$



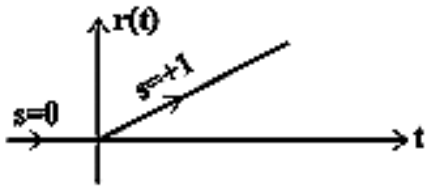


Fig. 1.37

Important observation:

1. r(t) = tu(t)
   r(at) = at u(at) = at u(t) = ar(t)
2. $r(at+b) = (at+b)u(at+b)$
   $= (at+b)u\left(t+\frac{b}{a}\right)$
   $= a\left(t+\frac{b}{a}\right)u\left(t+\frac{b}{a}\right) = ar\left(t+\frac{b}{a}\right)$
3. $r(-at+b) = (-at+b)u(-at+b)$
   $= (-at+b)u\left(-t+\frac{b}{a}\right)$
   $= a\left(-t+\frac{b}{a}\right)u\left(-t+\frac{b}{a}\right) = ar\left(-t+\frac{b}{a}\right)$

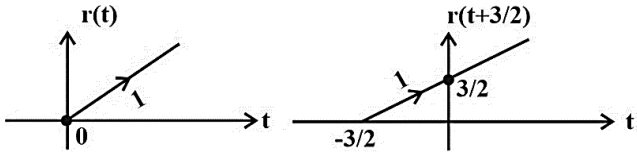
Impulse	Step	Ramp
$\delta(at) = \frac{1}{\lvert a\rvert}\delta(t)$	$u(at) = u(t)$	$r(at) = ar(t)$
$\delta(-at) = \frac{1}{\lvert -a\rvert}\delta(t)$	$u(-at) = u(-t)$	$r(-at) = ar(-t)$
$\delta(at+b) = \frac{1}{\lvert a\rvert}\delta\left(t+\frac{b}{a}\right)$	$u(at+b) = u\left(a\left(t+\frac{b}{a}\right)\right) = u\left(t+\frac{b}{a}\right)$	$r(at+b) = r\left(a\left(t+\frac{b}{a}\right)\right) = ar\left(t+\frac{b}{a}\right)$
$\delta(-at+b) = \frac{1}{\lvert -a\rvert}\delta\left(t-\frac{b}{a}\right)$	$u(-at+b) = u\left(-t+\frac{b}{a}\right)$	$r(-at+b) = r\left(a\left(-t+\frac{b}{a}\right)\right) = ar\left(-t+\frac{b}{a}\right)$

Table 1.2

Note: When time is scaled by factor of 'a', area of impulse signal gets multiplied by '|a|' factor, no change in step signal, whereas the slope of ramp signal gets multiplied by 'a' factor.

Example 1.17: Plot $r(2t+3) = 2r\left(t+\frac{3}{2}\right)$

Solution:



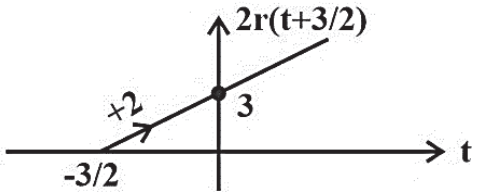


Fig. 1.38

Concept of slope:

$\Delta y = m \Delta x$

Slope: known

If Δx is covered on horizontal axis then $m\Delta x$ is covered on vertical axis.

Synthesis and analysis:

Synthesis:

Step 1 → Identify the BP and width between two BP.

Step 2→ Add subtract slope at every BP.

Example 1.18: Plot $x(t) = +2r(t+2) - 4r(t) + 2r(t-2)$

Solution:

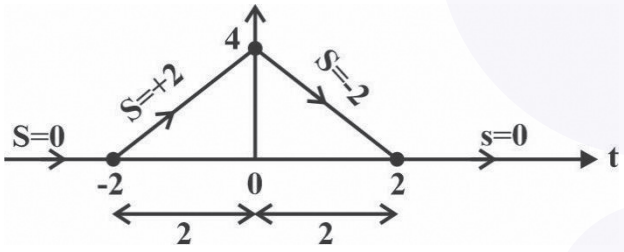


Fig. 1.39

Example 1.19:

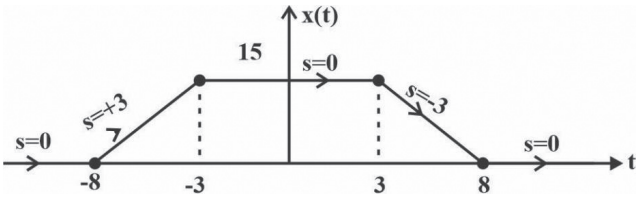


Fig. 1.40

Solution:

$$x(t) = 3r(t+8) - 3r(t+3) - 3r(t-3) + 3r(t-8)$$

Note: Coefficients of t is positive, so tracking will be from left to right.

Mixed Analysis and Synthesis

Types of BP:

Type 1:

1. BP at which vertical line is present.
2. ⊙
3. x(t) = Step change at BP [u(t-BP)]

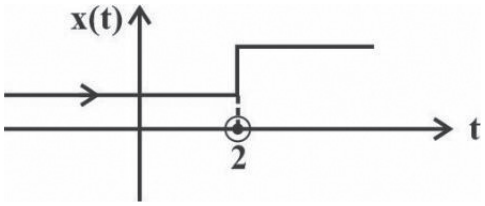


Fig. 1.41

Type 2:

1. BP around which slope changes
2. ⊡
3. x(t) = {Slope change around BP} [r(t-BP)]

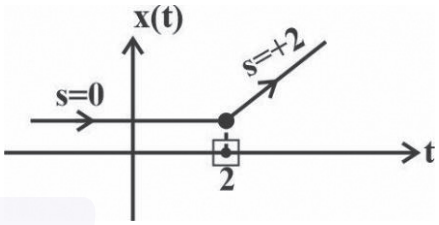


Fig. 1.42

Analysis:

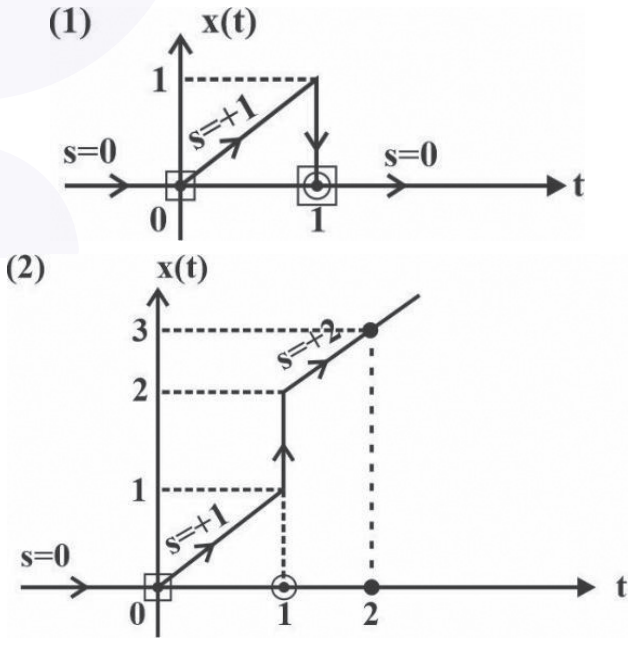


Fig. 1.43

Note: Slope change and height change at every breakpoint are to be noted properly.

Example 1.20:

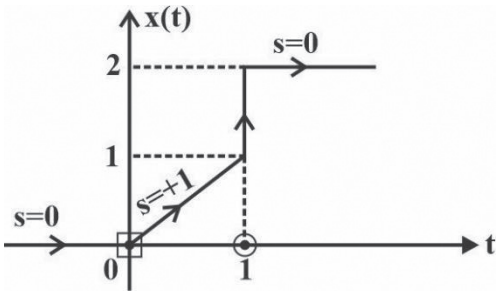


Fig. 1.44

Solution:

$$x(t) = 1 \cdot r(t-0) + u(t-1) - r(t-1)$$

Synthesis:

Example 1.20: $x(t) = r(t) + u(t-1)$

Solution:

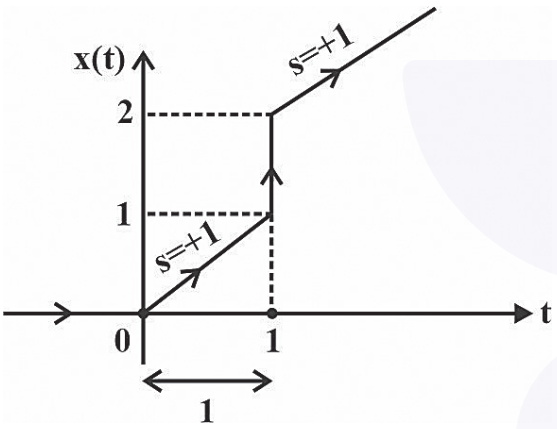


Fig. 1.45

Note: Unit step signals do not change direction of flow.

Parabolic signal:

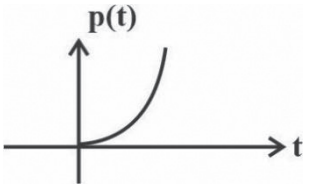


Fig. 1.46

$$p(t) = \frac{t^2}{2} u(t)$$

$$p(t) \xrightarrow{d/dt} r(t) \xrightarrow{d/dt} u(t) \xrightarrow{d/dt} \delta(t)$$

$$\delta(t) \xrightarrow{\int_{-\infty}^{t}} u(t) \xrightarrow{\int_{-\infty}^{t}} r(t) \xrightarrow{\int_{-\infty}^{t}} p(t)$$

Gate Pulse or Rectangular Pulse

1. Symmetric about vertical axis.
2.

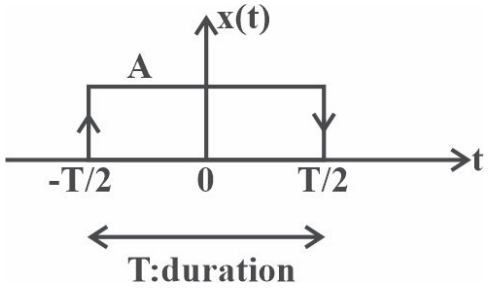


Fig. 1.47

3. $x(t) = \begin{cases} A & : |t| \le \frac{T}{2} = \frac{-T}{2} \le t \le \frac{T}{2} \\ 0 & : \end{cases}$

4. $x(t) = Au\left(t + \frac{T}{2}\right) - Au\left(t - \frac{T}{2}\right)$

5. $x(t) = A \text{ rect}\left(\frac{t}{T}\right) = A\Pi\left(\frac{t}{T}\right)$

Triangular pulse:

1. Even (symmetric about vertical axis).
2.

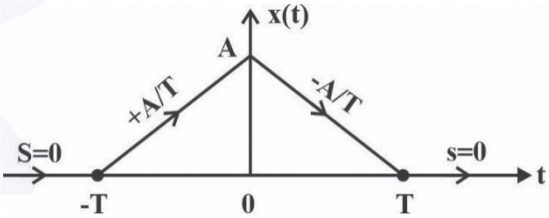


Fig. 1.48

3. $x(t) = \begin{cases} A\left(1 - \frac{|t|}{T}\right) & : |t| \le T : -T \le t \le T \\ 0 & : \text{elsewhere} \end{cases}$

4. $x(t) = \begin{cases} A(1 + t/T) & -T \le t < 0 \\ A & t = 0 \\ A(1 - t/T) & 0 < t \le T \end{cases}$

5. $x(t) = \frac{A}{T} r(t+T) - \frac{2A}{T} r(t) + \frac{A}{T} r(t-T)$

6. $A \text{ tri}\left(\frac{t}{T}\right) = A\,\Delta\left(\frac{t}{T}\right)$

Note:

In case of rectangular pulse, in the denominator of the argument, total width of the pulse is to be written, whereas in case of triangular pulse, half of the total width is to be written.

Sinc Function

1. $\text{Sinc}(t) = \dfrac{\sin \pi t}{\pi t}$
2. $\text{Sinc}(at) = \dfrac{\sin \pi (at)}{\pi (at)}$
3. $\dfrac{\text{Sin at}}{bt} = \dfrac{a}{b}\text{sinc}\left(\dfrac{at}{\pi}\right)$

Properties of sinc function:

1. $\lim\limits_{t\to 0}\text{sinc}(t) = \lim\limits_{t\to 0}\dfrac{\sin \pi t}{\pi t} = 1$
2. $\lim\limits_{t\to \pm\infty}\text{sinc}(t) = \lim\limits_{t\to \pm\infty}\dfrac{\sin \pi t}{\pi t} = \dfrac{Finite}{\infty} = 0$
3. $\text{sinc}(-t) = \dfrac{\sin \pi (-t)}{\pi (-t)} = \text{sinc}(t) \rightarrow$

   Symmetrical about vertical axis = even
4. Zero crossover time instances:

   $\text{sinc}(t) = 0$

   $\dfrac{\sin \pi t}{\pi t} = 0$

   $\sin \pi t = 0$

   $\sin \pi t = \sin n\pi$

   $\pi t = n\pi$

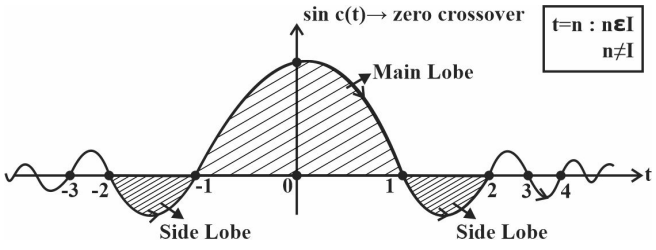
   $t = n \rightarrow n \in I \text{ and } n \neq 0$
5. 

   Fig. 1.49
6. $\int_{-\infty}^{\infty}\text{sinc}(t)dt = 1$
7. $\int_{-\infty}^{\infty}\text{sinc}^2(t)dt = 1$
8. $\int_{-\infty}^{\infty}\text{sinc}(kt)dt = \dfrac{1}{k}$
9. $\int_{-\infty}^{\infty}\text{sinc}^2(kt)dt = \dfrac{1}{k}$

Example 1.22: $X(\omega) = \dfrac{\sin 4\omega}{\omega}$,

calculate zero crossover frequencies = ?

Solution:

$X(\omega) = 4\,\text{sinc}\left(\dfrac{4\omega}{\pi}\right)$

Zero crossover frequencies

$\dfrac{4\omega}{\pi} = n$

$\omega = \dfrac{\pi n}{4}$

$2\pi f = \dfrac{n\pi}{4}$

$f = n/8 \rightarrow n \in I \text{ and } n \neq 0$

Sampling function:

$Sa(t) = \dfrac{\sin t}{t}$

$Sa(kt) = \dfrac{\sin kt}{kt}$

$\dfrac{\sin at}{bt} = \dfrac{a}{b}Sa(at)$

Properties:

1. $\lim\limits_{t\to 0}Sa(t) = \lim\limits_{t\to 0}\dfrac{\sin t}{t} = 1$
2. $\lim\limits_{t\to \pm\infty}Sa(t) = \lim\limits_{t\to \pm\infty}\dfrac{\sin t}{t} = 0$
3. $Sa(-t) = \dfrac{\text{Sin}(-t)}{(-t)} = Sa(t)$
4. Zero crossover time:

   $x(t) = \dfrac{\text{Sin}\, t}{t} = \text{Sinc}\left(\dfrac{t}{\pi}\right)$

   $t = n\pi \;\; n \in I, \;\; n \neq 0$

5.

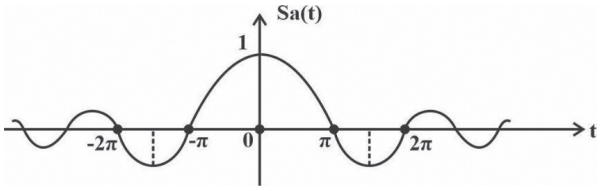


Fig. 1.50

6. $\int_{-\infty}^{\infty} Sa(t)\,dt = \int_{-\infty}^{\infty} \text{Sinc}\left(\frac{t}{\pi}\right)dt = \frac{1}{k} = \pi$

7. $\int_{-\infty}^{\infty} Sa^2(t)\,dt = \int_{-\infty}^{\infty} \text{Sinc}^2\left(\frac{t}{\pi}\right)dt = \frac{1}{k} = \pi$

Greatest Integer Function (GIF) or Floor Function

$[\cdot] \rightarrow GIF$

$x(t) = [t]$

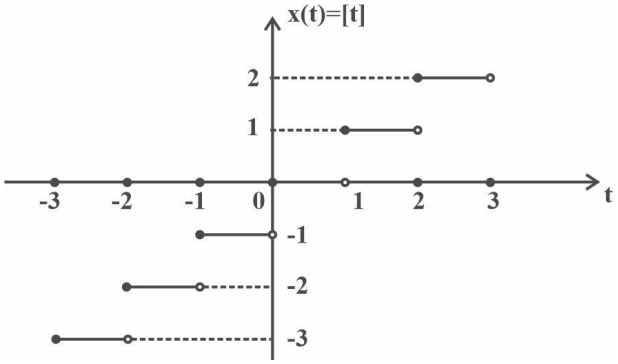
$$= \begin{cases} \text{Immediate previous Integer} & t : \textit{Fraction} \\ \text{Integer, itself} & t : \textit{Integer} \end{cases}$$

$[2 \cdot 3] = 2$

$[-2 \cdot 3] = -3$

$[2] = 2$

$$x(t) = [t] = \begin{cases} x(t) = [t] & t \\ -2 & : -2 \le t < -1 \\ -1 & : -1 \le t < 0 \\ 0 & : 0 \le t < 1 \\ 1 & : 1 \le t < 2 \\ 2 & : 2 \le t < 3 \end{cases}$$



$$x(t) = \begin{cases} x(t) = t - [t] & t \\ t+2 & : -2 \le t < -1 \\ t+1 & : -1 \le t < 0 \\ t-0 & : 0 \le t < 1 \\ t-1 & : 1 \le t < 2 \\ t-2 & : 2 \le t < 3 \end{cases}$$

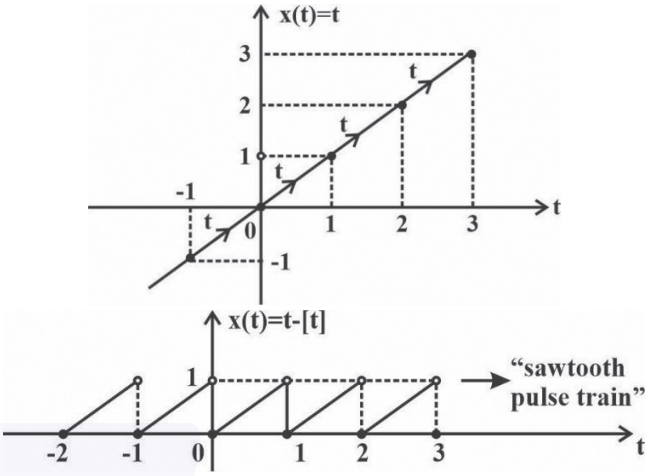


Fig. 1.51

Least Integer Function (LIF) or Ceiling Function

$[\cdot] \rightarrow LIF$

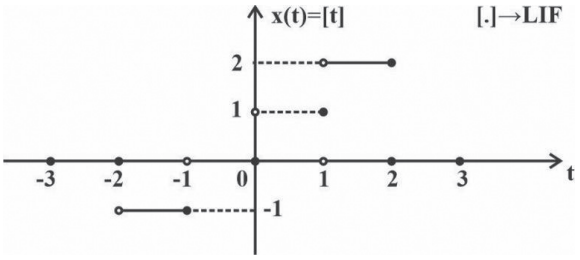
$$[t] = \begin{cases} \text{Immediate next Integer} & : \\ t \rightarrow \text{Non Integer} & \\ \text{Integer Itself} & : \\ t \rightarrow \text{Integer} & \end{cases}$$

$[2 \cdot 3] = 3$

$[-2 \cdot 3] = -2$

$[2] = 2$

$$x(t) = \begin{cases} x(t) = [t] & t \\ -1 & : -2 < t \le -1 \\ 0 & : -1 < t \le 0 \\ 1 & : 0 < t \le 1 \\ 2 & : 1 < t \le 2 \\ 3 & : 2 < t \le 3 \end{cases}$$



$$x(t)=\begin{cases} x(t)=t-[t] & [t]\rightarrow LIF \\ t+1 & :-2<t\le -1 \\ t+0 & :-1<t\le 0 \\ t-1 & :0<t\le 1 \\ t-2 & :1<t\le 2 \\ t-3 & :2<t\le 3 \end{cases}$$

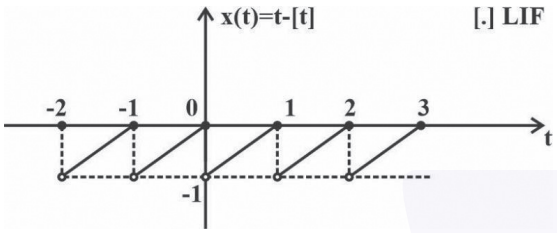


Fig. 1.52

Important points:

1. GIF → Floor function [t] = $\lfloor t \rfloor$
2. LIF → Ceiling function [t] = $\lceil t \rceil$
3. $x(t)=t-[t]$ is called as fractional part of $t=\{t\}$
4. $(t-\lfloor t \rfloor)=1+(t-\lceil t \rceil)$

Signum function:

1. $Sgn(t)=\begin{cases}-1 & :t<0\\ 0 & :t=0\\ 1 & :t>0\end{cases}=\begin{cases}\frac{|t|}{t} \text{ or } \frac{t}{|t|} & : t\neq 0\\ 0 & : t=0\end{cases}$
2. Nested: $Sgn\left[Sgn\left(Sgn(t)\right)\right]=Sgn(t)$

$$Sgn\left(Sgn(t)\right)=\begin{cases}-1: & t<0\\ 0 & t=0=Sgn(t)\\ 1 & t>0\end{cases}$$

3. $Sgn(t)=2u(t)-1=u(t)-u(-t)$

Example 1.23: $Sgn(\sin t)=\begin{cases}-1 & \sin t<0\\ 0 & \sin t=0\\ 1 & \sin t>0\end{cases}$

Solution:

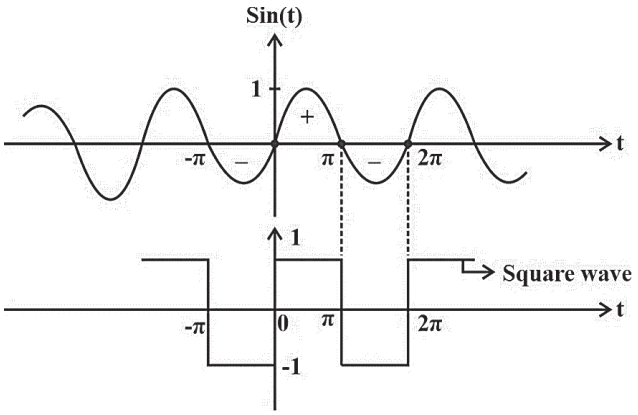


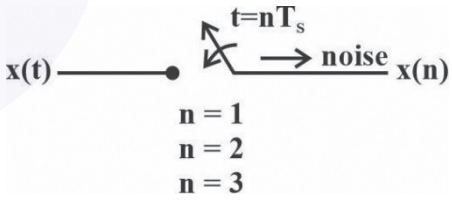
Fig. 1.53

Discrete-Time Signal (DTS)

1. Natural DTS (Not sampled version of CTS)
2. Converted DTS (Sampled version of CTS)

Symbol:

$x[n]$ vs n where $n\in I$



$$\begin{cases} x(n)=0 & : \text{when } n\notin I \\ x(n)=undefined & : \text{when } n\notin I \end{cases}$$

Representation:

1. Mathematical form: $x(n)=\left(\frac{1}{2}\right)^n \quad n\in I$
2. Sequenceform: $x(n)=\left\{....4,2,\underset{\uparrow}{1},\frac{1}{2},\frac{1}{4},\frac{1}{8}....\right\}$

   Sequence form
3. Graphical form:

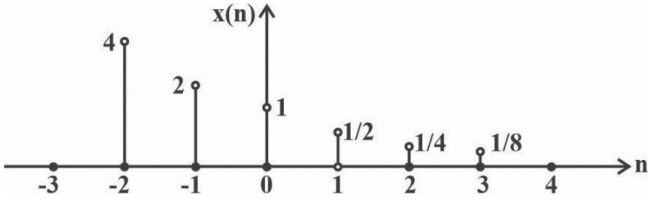


Fig. 1.54

Example 1.24: $x(n)=2(1)^n$

Solution:

$$x(n) = \left\{2, 2, 2, 2, \underset{\uparrow}{2}, 2\ldots\right\}$$
$$n=0$$

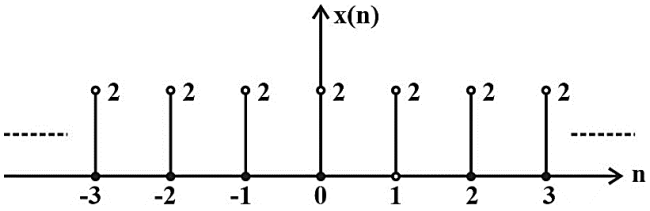


Fig. 1.54

Important points:

1. $x(n) = \left\{\underset{\uparrow}{1}, 2, 3\right\}$ → Finite duration
2. $x(n) = \left\{\underset{\uparrow}{1}, 2, 3\ldots\ldots\ldots\ldots\ldots\right\}$
   → *Infinite* duration
   → Right-sided Signal
3. $x(n) = \left\{\ldots\ldots\ldots\ldots 3, 3, 2, \underset{\uparrow}{1}\right\}$
   → *Infinite* duration
   → Left-sided Signal
4. $x(n) = \left\{\ldots\ldots\ldots\ldots 3, 2, 1, 4, 5\ldots\ldots\ldots\ldots\right\}$
   → *Infinite* duration
   ↑
   → Both-sided Signal

   → Can't comment on the default position of arrow

Note:

1. Default position of arrow for the RSS is at the first sample point from the left side.
2. Default position of the arrow for the LSS is at the last sample point at the right side.
3. When the signal is both-sided, we cannot comment on the default position of the arrow.

Operations on DTS

Amplitude scaling:

Given: x(n) vs n

Plot: A x(n) vs n

Example:

$$x(n) = \left\{1, \underset{\uparrow}{2}, 3\right\}$$
$$y(n) = 2x(n) = \left\{2, \underset{\uparrow}{4}, 6\right\}$$

Amplitude reversal:

Given: x(n) vs n

Plot: -x(n) vs n

Example:

$$x(n) = \left\{1, \underset{\uparrow}{2}, 3\right\}$$
$$y(n) = -x(n) = \left\{-1, \underset{\uparrow}{-2}, -3\right\}$$

Addition/Subtraction/Multiplication of signals:

Note: Perform operations between samples at same position.

Example:

$$x_1(n) = \left\{\begin{matrix} 1 & 2 & 3 \\ \uparrow & & \\ n=0, & n=1, & n=2 \end{matrix}\right\}$$

$$x_2(n) = \left\{\begin{matrix} 1 & 2 & 3 \\ & \uparrow & \\ n=-1 & n=0 & n=1 \end{matrix}\right\}$$

$$\begin{array}{cccc} 0 & 1 & 2 & 3 \\ & \uparrow & & \\ 1 & 2 & 3 & 0 \\ & \uparrow & & \\ \hline 1 & 3 & 5 & 3 \end{array}$$

$$x_1(n) + x_2(n) = \left\{1, \underset{\uparrow}{3}, 5, 3\right\}$$

Operation on independent variable:

Note: Whatever operation is performed, it will be strictly on the independent variable 'n'.

Time shifting:

Given: x(n) vs n

Plot: $x(n - n_0)$ vs n → Left shift arrow

$x(n + n_0)$ vs n → Right shift arrow

Note: $n_0 \in I$

Example:

$$x(n) = \{1, 2, 3, \underset{\uparrow}{1}, 1, 1, \ldots\ldots\}$$

$$x(n-1) = \{1, 2, \underset{\uparrow}{3}, 1, 1, 1, \ldots\ldots\}$$

$$x(n-3) = \{\underset{\uparrow}{1}, 2, 3, 1, 1, 1, \ldots\ldots\}$$

$$x(n-4) = \{\underset{\uparrow}{0}, 1, 2, 3, 1, 1, 1, \ldots\ldots\}$$

Time reversal:

Given: x(n) vs n

Plot: x(-n) vs n

S.1→ Make x(n) an equivalent signal.

S.2→ Exchange symmetric elements.

Example:

$$x(n) = \{1, \underset{\uparrow}{4}, 5, -2\} = \{0, 1, \underset{\uparrow}{4}, 5, -2\}$$

$$x(-n) = \{-2, 5, \underset{\uparrow}{4}, 1, 0\}$$

Time scaling:

Given: x(n) vs n

Plot: x(an) vs n

A should be rational: $\left(\frac{Integer}{Integer}\right)$

Example:

Case 1:

$a > 1$

$$x(n) = \begin{matrix} 1, & 2, & 3, & -1, & 4, & 5, & 6, & 3, & -2, & 10, & 5 \\ & & & \uparrow & & & & & & & \end{matrix}$$

$$x(2n) = \begin{matrix} 1, & 2, & 3, & -1, & 4, & 5, & 6, & 3, & -2, & 10, & 5 \\ & & & \uparrow & & & & & & & \\ \frac{-3}{2} & \frac{-2}{2} & \frac{-1}{2} & \frac{0}{2} & \frac{1}{2} & \frac{2}{2} & \frac{3}{2} & \frac{4}{2} & \frac{5}{2} & \frac{6}{2} & \frac{7}{2} \end{matrix}$$

$$x(2n) = \begin{matrix} 2, & -1, & 5, & 3, & 10 \\ & \uparrow & & & \\ -1 & n=0 & 1 & 2 & 3 \end{matrix}$$

Above operation is known as decimation in time domain.

Case 2: $x(n) = \begin{matrix} 1, & 2, & 3, & 1, & 5 \\ & \uparrow & & & \end{matrix}$

$$x\left(\frac{n}{2}\right) = \begin{matrix} 1, & 2, & 3, & 1, & 5 \\ & \uparrow & & & \\ -\frac{1}{1/2} & n=\frac{0}{1/2} & \frac{1}{1/2} & \frac{2}{1/2} & \frac{3}{1/2} \end{matrix}$$

$$x\left(\frac{n}{2}\right) = \begin{matrix} 1, & 0, & 2, & 0, & 3, \\ & n=-1 & \uparrow & n=1 & \\ n=-2 & & & & n=2 \\ 0, & 1, & 0, & 5, & \\ n=3 & & n=5 & & \\ & n=4 & & n=6 & \end{matrix}$$

Above operation is known as interpolation in time domain.

Example:

$$x(n) = \begin{matrix} 1, & 2, & -1, & -1, \\ & & \uparrow & \\ \frac{-2}{3/4} & \frac{-1}{3/4} & \frac{n=0}{3/4} & \frac{1}{3/4} \\ 3, & 2, & -1, & -1, \\ \frac{1}{3/4} & \frac{2}{3/4} & \frac{3}{3/4} & \end{matrix}$$

$$a = \frac{3}{4}$$

$$x\left(\frac{3n}{4}\right) = \begin{matrix} -1, & 0, & 0, & 0, & 5 \\ \uparrow & n=1 & n=2 & n=3 & \\ n=0 & & & & n=4 \end{matrix}$$

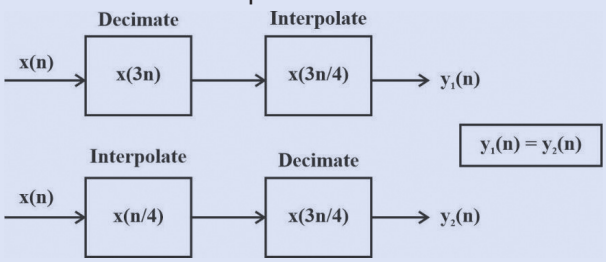
Note:

$x(n) = \{1\}$ (↑ $n=0$)

$x(2n) = \{1\}$ (↑)

$x(3n) = \{1\}$ (↑)

Note: Decimation do not always lead to reduction in samples.



Note: $x(n) = \{1, \underset{\uparrow}{2}, 3, 4\}$

$a = \sqrt{2} \rightarrow$ *Irrational*

$a = \pi$

$x(\sqrt{2}n)$

$a = (\pi n)$ *Invalid* operation

Mixed operation:

Given: x(n) vs n

Plot: x(-3n + 2) vs n

Natural order : $x(n) \rightarrow x(n+2) \rightarrow x(3n+2) \downarrow x(-3n+2)$

Other order : $x(n) \rightarrow x(3n) \rightarrow x\left(3\left(n+\frac{2}{3}\right)\right)$

$\rightarrow$ Invalid

Note: In case of DTS, the natural order is performed strictly.

Example 1.25: $x(n) = \{1, 1, \underset{\uparrow}{2}, 3, -1, 4, 5\}$, $x(-2n+3)$?

$$x(n+3) = \begin{Bmatrix} 1, & 1, & 2, & 3, & -1, & 4, & 5 \\ \frac{-5}{2} & \frac{-4}{2} & \frac{-3}{2} & \frac{-2}{2} & \frac{-1}{2} & \underset{\uparrow}{\frac{n=0}{2}} & \frac{1}{2} \end{Bmatrix}$$

$$x(2n+3) = \begin{Bmatrix} 1 & 3 & 4 \\ -2 & -1 & \underset{\uparrow}{n=0} \end{Bmatrix} \rightarrow \begin{Bmatrix} 1, & 3, & 4, & 0, & 0 \\ & & \uparrow & & \end{Bmatrix}$$

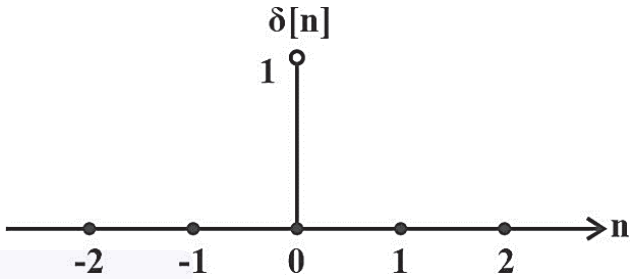
$$x(-2n+3) = \begin{Bmatrix} 0, & 0, & 4 & 3, & 1 \\ & & \uparrow & & \end{Bmatrix} \rightarrow \begin{Bmatrix} 4, & 3, & 0 \\ \uparrow & & \end{Bmatrix}$$

Standard DT Signal

Unit impulse sequence:

Symbol: $\delta[n]$

Graph:



Definition:

$$\delta[n] = \begin{cases} 1 & : \ n = 0 \\ 0 & : \ n \neq 0 \end{cases}$$

$\delta[n] = \{1\}$ (↑)

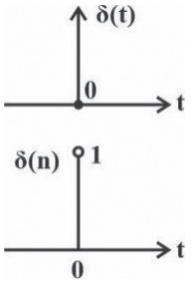
Note: $\delta[0] = 1$

$\delta(t)$ vs $\delta(n)$

1. $\delta(n)$ is not sampled version of $\delta(t)$

2.

Parameter	$\delta(t)$	$\delta(n)$
Duration →	Finite	Finite
Amplitude at B.P. (t=0/n=0) →	∞	1



3. $\delta(at) = \frac{1}{|a|}\delta(t)$ but $\delta(an) = \delta(n)$

4. $\delta(-at+b)=\delta\left(-a\left(t-\frac{b}{a}\right)\right)$

$$=\frac{1}{|a|}\delta\left(t-\frac{b}{a}\right)$$

but $\delta(-an+b)=\delta\left(-a\left(n-\frac{b}{a}\right)\right)$

$$=\delta\left(n-\frac{b}{a}\right)$$

Properties:

1. $\delta(-n)=\delta(n)$

2. $\delta(an)=\delta(n)$

3. $\delta(-an+b)=\delta\left(-a\left(n-\frac{b}{a}\right)\right)=\delta\left(n-\frac{b}{a}\right)$

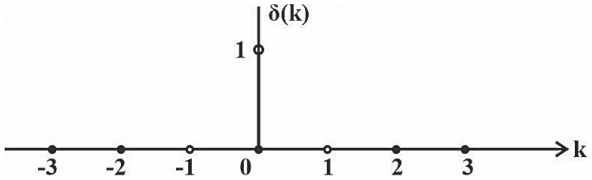
Note: In case of time scaling property of impulse signal in DTS, the denominator does not get scaled by a factor of 'a' because it is not an approximated signal. It is obtained from sampling CTS impulse signal.

4. $x(n)\delta(n)=x(0)\delta(n)$

5. $x(n)\delta(-an+b)=x\left(\frac{b}{a}\right)\delta\left(n-\frac{b}{a}\right)$

6. $\sum_{k=-\infty}^{\infty}\delta(k)=\ldots\delta(-3)+\delta(-2)+\delta(-1)+\delta(0)+\delta(1)+\delta(2)+\delta(3)+\ldots$

$=1$



7. $\sum_{n=k_1}^{n=k_2}x(n)\delta(-an+b)=\sum_{n=k_1}^{n=k_2}x\left(\frac{b}{a}\right)\delta\left(n-\frac{b}{a}\right)$

$$=\begin{cases}x\left(\frac{b}{a}\right) & \text{is impulse falls b/w } k_1\le n\le k_2\\ 0 & \text{otherwise}\end{cases}$$

8. $\delta(n)\times\delta(n)\times\ldots\ldots\infty \quad \text{times}=\delta(n)$

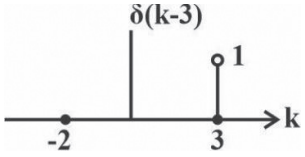
9. $\delta(n)+\delta(-n)=2\delta(n)$

10. $\delta(n)-\delta(-n)=0$

Example 1.26: $\sum_{k=-\infty}^{\infty}\delta(2k-6)=\ ?$

Solution: $\sum_{k=-\infty}^{\infty}\delta(2k-6)=\sum_{k=-\infty}^{\infty}\delta(2(k-3))$

$$=\sum_{k=-\infty}^{\infty}\delta(k-3)=1$$



Synthesis and analysis:

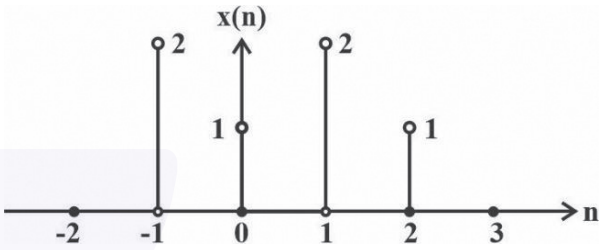


Fig. 1.55

1. $x(n)=\left\{\begin{matrix}2, & \underset{\uparrow}{1}, & 2, & 1\\ n=1 & n=0 & n=1 & n=2\end{matrix}\right\}$

2. $x(n)=2\delta(n+1)+\delta(n)+2\delta(n-1)+1\delta(n-2)$

$$x(n)=\sum_{k=-\infty}^{\infty}x(k)\delta(n-k) \qquad n,k\in I$$

Every DT signal can be written as weighted sum of impulse and shifted impulses.

$$x(n)=3\delta(n+3)+2\delta(n+1)+\delta(n)-4\delta(n-2)$$

$$x(n)=\left\{\begin{matrix}3, & 0, & 2,\\ n=-3 & n=-2 & n=-1\\ \underset{\uparrow}{1}, & 0, & 0,\\ n=0 & n=1 & n=2\end{matrix}\right\}$$

Unit step sequence: Sampled version of u(t)

Graph:

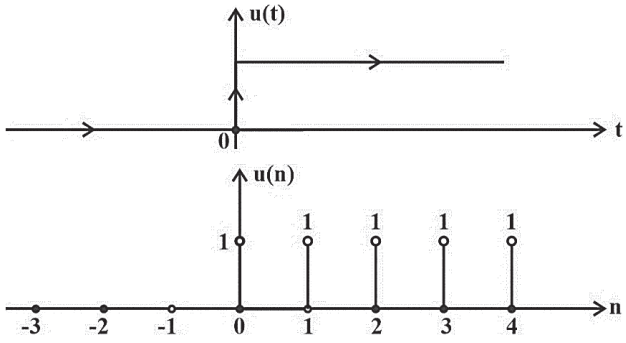


Fig. 1.56

Definition:

$$u(n)=\begin{cases}1 & : \ n\geq 0\\ 0 & : \ n<0\rightarrow n\leq -1\end{cases}$$

$$u(n)=\begin{cases}(1)^n & : \ n\geq 0\\ 0 & : \ n<0; n\leq -1\end{cases}$$

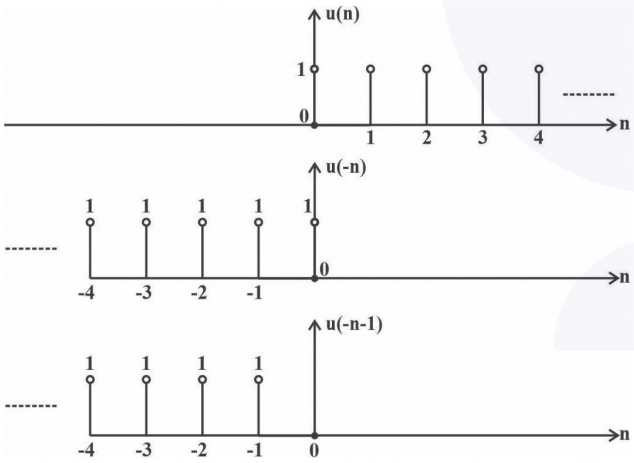


Fig. 1.57

1. $u(n)+u(-n)=1+\delta(n)=(1)^n+\delta(n)$
2. $u(n)+u(-n-1)=1=(1)^n \xleftrightarrow{\text{Analogy}}$
   $u(t)+u(-t)=1$
3. $u(n)\cdot u(-n)=\delta(n)$
4. $u(n)\cdot u(-n-1)=0 \xleftrightarrow{\text{Analogy}}$
   $u(t)\cdot u(-t)=0$

   Analogous of u(-t) is u(-n-1)

Note:

u(n) + u(-n - 1) = 1

Hence

$n\rightarrow n-4$

$$u(n-4)+u(-(n-4)-1)=1$$

$$u(n-4)+u(-n+3)=1$$

Relation between δ(n) and u(n):

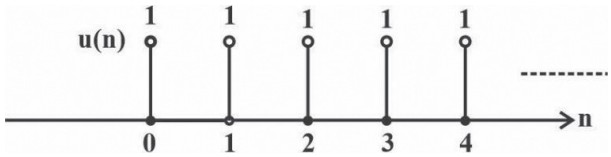


Fig. 1.58

1. $\delta(n)=u(n)-u(n-1)$
2. $u(n)=\delta(n)+\delta(n-1)+\delta(n-2)+.....$

Important

$$u(n)=\sum_{k=0}^{\infty}\delta(n-k)$$

$n-k=p$

$at\ k=0\rightarrow p=n$

$at\ k=\infty\rightarrow p=-\infty$

$$u(n)=\sum_{p=n}^{-\infty}\delta(p)=\sum_{p=-\infty}^{n}\delta(p)$$

$$u(n)=\sum_{k=-\infty}^{n}\delta(k)$$

Note:

$$u(n)=\sum_{k=0}^{\infty}\delta(n-k)$$

$$u(n)=\sum_{k=-\infty}^{n}\delta(k)$$

Observation:

$$\sum_{k=0}^{¥}ä(n-k)=u(n)$$

$$\sum_{k=5}^{\infty}\delta(n-2-k)=u(n-7)$$

$$\sum_{k=-\infty}^{n}\delta(k)=u(n)$$

$$\sum_{k=-\infty}^{n+5}\delta(k-2)=u(n+3)$$

Example 1.27:

Given $x(n) = 1 - \sum_{k=4}^{\infty} \delta(n-k-1) = u(Mn-N)$, calculate M and N?

Solution:

$$x(n) = 1 - u(n-5) = u[Mn-4] = u(-n+4)$$

$$u[n] + u[-n-1] = 1$$

$$n \to n-5$$

$$u(n-5) + u(-n+4) = 1$$

$$1 - u(n-5) = u(-n+4)$$

M = -1 and N = -4

Synthesis and analysis:

Analysis:

Case 1: Finite duration signal

$$x(n) = \{1, \underset{\uparrow}{3}, 4, 6\}$$

$$x(n) = \begin{Bmatrix} 0, & 1, & 3, & 4, & 6, & 0 \\ & & \uparrow & & & \\ -2 & -1 & n=0 & 1 & 2 & 3 \end{Bmatrix}$$

Left to right

$$x(n) = u(n+1) + 2u(n) + u(n-1) + 2u(n-2) - 6u(n-3)$$

Note: At any time instant, (Right sample-Left sample) u(n-BP) is done in case of RSS and (Left sample-Right sample)u(-n+BP) is done in case of LSS.

Case 2: Infinite duration → Right sided signal

$$x(n) = \{3, -1, \underset{\uparrow}{2}, -2, 4, 4, 4, \ldots\ldots\}$$

$$x(n) = \begin{Bmatrix} 0, & 3, & -1, & 2, & -2, & 4, & 4, & 4 & \ldots \\ & & & \uparrow & & & & & \\ -3 & -2 & -1 & n=0 & 1 & 2 & 3 & 4 & \end{Bmatrix}$$

Left to right

$$x(n) = 3u(n+2) - 4u(n+1) + 3u(n) - 4u(n-1) + 6u(n-2)$$

Case 3: Infinite duration + Left sided signal

$$x(n) = \{\ldots\ldots\ldots\ 4, \ 4, \ -2, \ \underset{\uparrow}{2}, \ 1, \ 3,\}$$

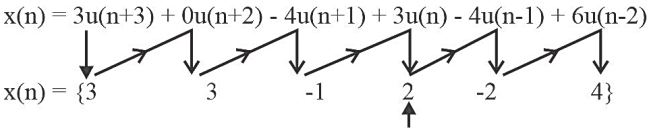
$$x(n) = \begin{Bmatrix} \ldots\ldots\ldots & 4, & 4, & -2, & 2, & 1, & 3, & 0 \\ & & & & \uparrow & & & \\ & -3 & -2 & -1 & n=0 & 1 & 2 & 3 \end{Bmatrix}$$

Right to left

$$x(n) = 3u(-n+2) - 2u(-n+1) + 1u(-n) - 4u(-n-1) + 6u(-n-2)$$

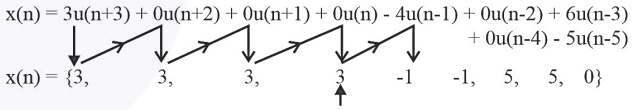
Synthesis:

1. $x(n) = 3u(n+3) - 4u(n+1) + 3u(n) - 4u(n-1) + 6u(n-2)$



$$x(n) = \{3, \ 3, \ -1, \ \underset{\uparrow}{2}, \ -2, \ 4, \ 4, \ 4, \ \ldots.\}$$

2. $x(n) = 3u(n+3) - 4u(n-1) + 6u(n-3) - 5u(n-5)$



Note:

1. Sign of n is same
2. Sum of amplitude coefficient = 0 [Finite duration + Finite amplitude signal]

Unit Ramp Seq

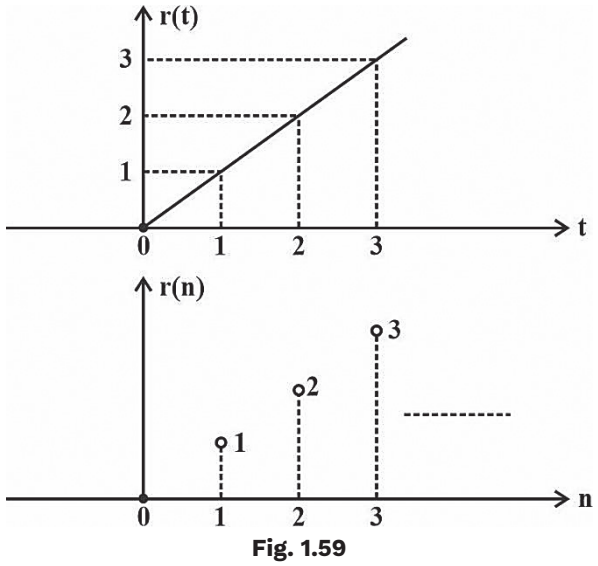


Fig. 1.59

$$x(n)=\left\{\begin{matrix}0, & 1, & 2, & 3, & 4 & \ldots\ldots & \infty\\ \uparrow & & & & & & \\ n=0 & 1 & 2 & 3 & 4 & \ldots\ldots & \end{matrix}\right\}$$

1. $r(n)=\delta(n-1)+2\delta(n-2)+3\delta(n-3)+4\delta(n-4)+\ldots$

2. $r(n)=u(n-1)+u(n-2)+u(n-3)+u(n-4)+\ldots$

$$r(n)=\sum_{k=0}^{\infty}u(n-k-1)$$

$n-k-1=p$

for $k=0\rightarrow p=n-1$

for $k=\infty\rightarrow p=-\infty$

$$r(n)=\sum_{p=n-1}^{-\infty}u(p)$$

$$r(n)=\sum_{p=-\infty}^{n-1}u(p)$$

$$r(n)=\sum_{k=-\infty}^{n-1}u(k)$$

Note:

1. $u(n)=\sum_{k=0}^{\infty}\delta(n-k)$

$$u(n)=\sum_{k=-\infty}^{n}\delta(k)$$

2. $r(n)=\sum_{k=0}^{\infty}u(n-k-1)\rightarrow r(n-1+1)$

$r(n)=\sum_{k=-\infty}^{n-1}u(k)\rightarrow r(n-1+1)$

3. $r(n)=nu(n)=nu(n-1)$

Example: $r(n)=(n+2)u(n+2)=(n+2)u(n+1)$

Even, Odd, Neither Even Nor Odd (NENO) Signals

Let $x(t)/x(n)\rightarrow$ Real, complex, and imaginary

Even signal:

$x(t)/x(n)$ is even if the following condition is satisfied.

$x(-t)=x(t)$

$x(-n)=x(n)$

Graph: Symmetric about vertical axis.

Examples:

$\delta(t)$

$\delta(n)$

$\cos t$

$\cos n$

$\sin c(t)$

Properties:

1. $\int_{-\infty}^{\infty}x(t)dt=2\int_{0}^{\infty}x(t)dt$

$=$ *may or may not be zero*

2. $\sum_{n=-\infty}^{\infty}x(n)\neq 2\sum_{n=0}^{\infty}x(n)=x(0)+2\sum_{n=1}^{\infty}x(n)$

Odd signal:

$x(t)/x(n)$: Odd signal, if the following condition is satisfied.

$x(-t)=-x(t)$

$x(-n)=-x(n)$

Graph:

Symmetric about origin.

Example:

$\sin t$

$sgn(t)$

t

$\frac{1}{t}$

n

$\sin n$

Properties:

1. $x(-t)=-x(t)\xrightarrow{t=0}2x(0)=0$

$x(-n)=-x(n)\xrightarrow{n=0}2x(0)=0$

$\rightarrow x[0]=0$

Provided $t=0/n=0$ is part of domain.

2. $\int_{-\infty}^{\infty}x(t)dt=0$

3. $\sum_{k=-\infty}^{\infty} x(k) = 0$

Neno: Not even + not odd

Example:

$u(t), u(n)$

$r(t), r(n)$

$\delta(n-3), \delta(t-3)$

Decomposition of signal:

$x(t) = x_e(t) + x_0(t)$	$x(n) = x_e(n) + x_0(n)$
$x_e(t) = \frac{x(t) + x(-t)}{2}$	$x_e(n) = \frac{x(n) + x(-n)}{2}$
$x_0(t) = \frac{x(t) - x(-t)}{2}$	$x_0(n) = \frac{x(n) - x(-n)}{2}$

Example 1.28: $x(t) = u(t),\ x_0(t) = ?$

$$x_e(t) = \frac{u(t) + u(-t)}{2} = \frac{1}{2} \rightarrow [Even]$$

$$x_0(t) = \frac{u(t) - u(-t)}{2} = \frac{1}{2} Sgn(t) \rightarrow [Odd]$$

Conjugate Symmetric and Conjugate Anti-Symmetric

Let $x(t) / x(n)$ is complex.

Conjugate symmetric:

1. Also known as even conjugate.

$$x(-t) = x^*(t)$$

$$x(-n) = x^*(n)$$

2. If x(t) is conjugate symmetric then
$\text{Re}\{x(t)\} \rightarrow$ Even
$\text{Im}\{x(t)\} \rightarrow Odd$

Example: $x(t) = e^{jt} \rightarrow$ Conjugate symmetric

$x(-t) = e^{-jt}$

$x^*(-t) = e^{jt}$

$x(t) = x^*(-t)$

$e^{jt} = \cos t + i \sin t$

$\text{Re}\{x(t)\} = \cos t \rightarrow Even$

$\text{Im}\{x(t)\} = \sin t \rightarrow Odd$

Conjugate anti-symmetric:

1. Odd conjugate

$$x(-t) = -x^*(t)$$

$$x(-n) = -x^*(n)$$

2. *If* $x(t)$ *is conjugate* anti symmetric then

$\text{Re}\{x(t)\} = \cos t \rightarrow Odd$

$\text{Im}\{x(t)\} = \sin t \rightarrow Even$

Example: $x(t) = te^{jt}$

$x(-t) = -te^{-jt}$

$x^*(-t) = -te^{jt}$

$x(t) = -x^*(-t)$

$x(t) =$ CAS.

$\text{Re}\{x(t)\} = t \cos t \rightarrow Odd$

$\text{Im}\{x(t)\} = t \sin t \rightarrow Even$

Even Conjugate (EC) part and Odd Conjugate (OC) part:

$\rightarrow x(t)$: complex	$\rightarrow x(n)$: complex
$x_{EC} = \frac{x(t) + x^*(-t)}{2}$	$x_{EC} = \frac{x(n) + x^*(-n)}{2}$
$x_{OC} = \frac{x(t) - x^*(-t)}{2}$	$x_{OC} = \frac{x(n) - x^*(-n)}{2}$

Example 1.29: $x(n) = \{1 + 2j,\ \underset{\uparrow}{4},\ 5 - j\}$

Solution:

$\sum_{n=-\infty}^{\infty} x_{EC}(n) =$

$\sum_{n=-\infty}^{\infty} x_{OC}(n) =$

$$x(n) = \{1+2j,\ \underset{\uparrow}{4},\ 5-j\}$$

$$x^*(n) = \{5+j,\ \underset{\uparrow}{4},\ 1-2j\}$$

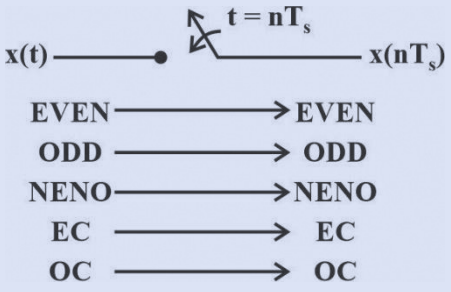
$$x_{EC}(n) = \{3+1.5j,\ \underset{\uparrow}{4},\ 3-1.5j\}$$

$$\sum_{n=-\infty}^{\infty} x_{EC}(n) = 10$$

$$x_{OC}(n) = \{-2+0.5j,\ \underset{\uparrow}{0},\ 2+0.5j\}$$

$$\sum_{n=-\infty}^{\infty} x_{OC}(n) = j$$

Note: Effect after sampling



Periodic and Non-Periodic Signal

CTS

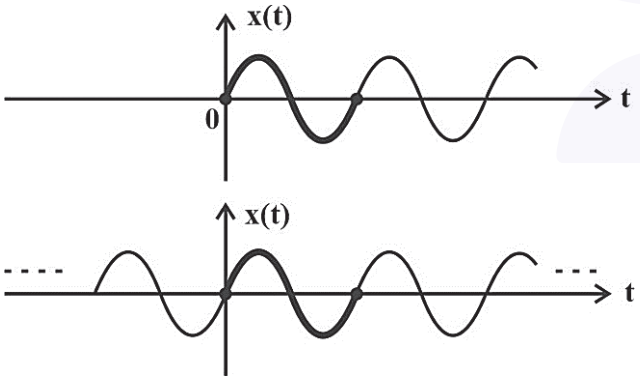


Fig. 1.60

A CTS is periodic if:

1. x(t) must have the smallest repeating assembly from -∞ to +∞.
2. $x(t + kT_0) = x(t)$

   $x(t + T) = x(t)$

   $T = kT_0 \rightarrow$ Time period $[k \in I]$

   $(T)_{\text{smallest}} = T_0 \rightarrow$ "Fundamental time period."

$$\frac{1}{T_0} = f_0 \rightarrow \text{Fundamental frequency}$$

Restrictions on T_0	*Allowed* T_0
1. $T_0 \neq 0$	1. T_0 = constant
2. $T_0 \neq \infty$	2. T_0 = Integer / *Non* Integer
3. $T_0 \neq -ve$	3. T_0 = Rational (p/q) / Irrational ($\sqrt{2}, \sqrt{3}, \pi$...)
4. $T_0 \neq f(t)$	

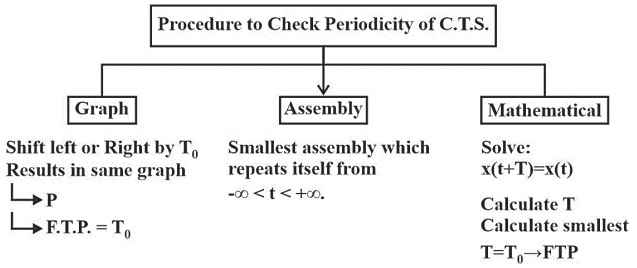


Fig. 1.61

30. Find fundamental time period of $x(t) = \sin t$.

Solution:

$$x(t+T) = x(t)$$

$$\sin(t+T) = \sin t = \sin(t + 2n\pi)$$

$$t + T = t + 2n\pi$$

$$T = 2n\pi \qquad \boxed{n \in I}$$

$\sin \alpha = \sin \beta$

$\alpha = \beta + 2n\pi$

$T0 = (T)_{\text{smallest}} = 2\pi$

F.T.P $T_0 = 2\pi$

31. Find fundamental time period of $x(t) = \sin t^2$

Solution:

$$x(t+T) = x(t)$$

$$\sin(t+T)^2 = \sin t^2$$

$$(t+T^2) = t^2 + 2n\pi$$

$$t^2 + T^2 + 2tT = t^2 + 2n\pi$$

$$T^2 + (2t)T - 2n\pi = 0$$

$$T = \frac{-2t \pm \sqrt{(2t)^2 + 8n\pi}}{2}$$

$$(T)_{smallest} = T_0 = f(t)$$

$x(t)$ is non-periodic

32. Find fundamental time period of $x(t) = \sin\sqrt{t}$.

Solution:

$x(t+T) = x(t)$

$\sin\left(\sqrt{t+T}\right) = \sin\left(\sqrt{t}\right)$

$\sqrt{t+t} = \sqrt{t} + 2n\pi$

$t + T = t + 4n^2\pi^2 + 4n\pi\sqrt{t}$

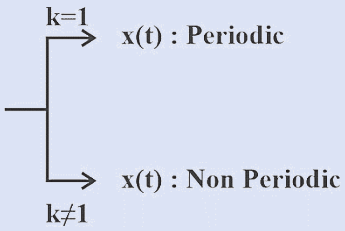
$T = 4n^2\pi^2 + 4n\pi\sqrt{t} = f(t) \rightarrow$ Non-periodic

Note:

$\mathbf{x(t) = A\cos\left(\omega_0 t^k + \phi\right)}$

OR

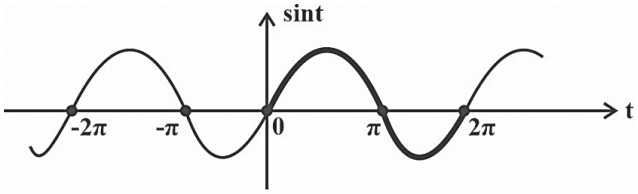
$\mathbf{A\sin\left(\omega_0 t^k + \phi\right)}$



Effects of operation:

Signal		F.T.P.
$x(t)$	$\rightarrow$	T_0
$x(t \pm t_0)$	$\rightarrow$	T_0
$-kx(t)$	$\rightarrow$	T_0
$x(t) \pm A$	$\rightarrow$	T_0
$x(-t)$	$\rightarrow$	T_0
$x(at)$	$\rightarrow$	$\frac{T_0}{\lvert a\rvert}$
$x(t)$	$\rightarrow$	T_0
$-Ax(-at+b)$	$\rightarrow$	$\frac{T_0}{\lvert a\rvert}$

Periodicity of sinusoidal signal:



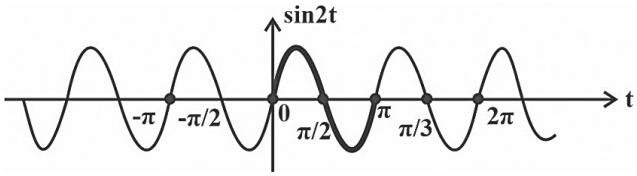


Fig. 1.62

Important:

$\sin 1t \rightarrow$ 1 Cycle in 2π duration → $T_0 = 2\pi$

$\sin 2t \rightarrow$ 2 Cycle in 2π duration → $T_0 = 2\pi/2$

$\sin\sqrt{2}t \rightarrow$ π Cycle in 2π duration → $T_0 = 2\pi/\pi$

$\sin\sqrt{2}t \rightarrow \sqrt{2}$ Cycle in 2π duration → $T_0 = 2\pi/\sqrt{2}$

$\sin\omega_0 t \rightarrow \omega_0$ Cycle in 2π duration → $T_0 = 2\pi/\omega_0$

$B + A\cos\left(\omega_0 t + \phi\right)$

Important results:

1. $A\cos\omega_0 t$ or $A\sin\omega_0 t$: $T_0 = 2\pi/\omega_0$
2. $A\cos\left(\omega_0 t + \phi\right)$ or $A\sin\left(\omega_0 t + \phi\right)$: $T_0 = 2\pi/\omega_0$
3. $B + A\cos\left(\omega_0 t + \phi\right)$ or $B + A\sin\left(\omega_0 t + \phi\right)$: $T_0 = 2\pi/\omega_0$

Effects of modulus operator:

Case 1:

Sinusoid free from dc addition

$x(t) = A\cos\left(\omega_0 t + \phi\right) \rightarrow T_0 = 2\pi/\omega_0$

$y(t) = \lvert x(t)\rvert = \lvert A\cos\left(\omega_0 t + \phi\right)\rvert \rightarrow T_0 = \pi/\omega_0$

$t \rightarrow t + \pi/\omega_0$

$= \lvert -A\cos\left(\omega_0 t + \phi\right)\rvert$

$= \lvert A\cos\left(\omega_0 t + \phi\right)\rvert$

Case 2:

Sinusoid having dc addition

$x(t) = A + B\cos\left(\omega_0 t + \phi\right) \rightarrow T_0 = 2\pi/\omega_0$

$y(t) = \lvert x(t)\rvert = \lvert A + B\cos\left(\omega_0 t + \phi\right)\rvert \rightarrow T_0 = \pi/\omega_0$

$t = t + \pi/\omega_0$

$\lvert A - B\cos\left(\omega_0 t + \phi\right)\rvert$

Effects of square:

$x(t) = A\cos(\omega_0 t + \phi) \rightarrow T_0 = 2\pi / \omega_0$

$y(t) = x^2(t) = A^2\cos^2(\omega_0 t + \phi)$

$$\boxed{\cos^2\theta = \frac{1+\cos 2\theta}{2}}$$

$$= \frac{A^2}{2} + \frac{A^2}{2}\cos(2\omega_0 t + 2\phi) \rightarrow T_0 = \frac{2\pi}{2\omega_0} = \frac{\pi}{\omega_0}$$

Periodicity of complex exponential:

$x(t) = e^{\pm j(\omega_0 t)} = \cos\omega_0 t \pm j\sin\omega_0 t \rightarrow T_0 = 2\pi / \omega_0.$

$x(t) = e^{\pm j(\omega_0 t + \phi)} \rightarrow T_0 = 2\pi / \omega_0$

Examples:

1. $\sin 2t \rightarrow T_0 = \frac{2\pi}{2} = \pi$

2. $\sin\sqrt{2}t \rightarrow T_0 = \frac{2\pi}{\sqrt{2}} = \pi\sqrt{2}$

3. $\sin\frac{6\pi}{5}t \rightarrow T_0 = \frac{2\pi}{6\pi} \times 5 = \frac{5}{3}$

Note:

$x(t) = \underset{P}{\sin t} \times \underset{NP}{u(t)} = NP$

$$x(t) = \underset{P}{\sin t} + \underset{NP}{u(t)} = \begin{cases} \sin t & +<0 \\ 1+\sin t & +>0 \end{cases} = NP$$

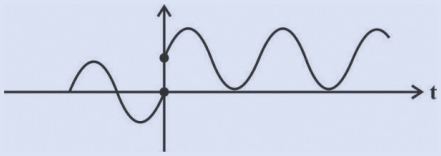


Table:

$x_2(t)$	$x_2(t)$	$x_1(t) + x_2(t)$	$x_1(t)x_2(t)$
P	*P*	?	?
P	*NP*	*NP*	*NP*
NP	*P*	*NP*	*NP*
NP	*NP*	*NP*	*NP*

Table 1.3

Note:

1. CT sinusoids and CT complex exponentials are always periodic.
   (When $\mathbf{t^k \rightarrow k = 1}$).
2. Linear combination of CT sinusoids and CT exponentials may or may not be periodic.

Periodicity of linear combination of CT sinusoids:

$$Given\ x(t) = A + \underbrace{B\cos(\omega_1 t + \phi_1)}_{T_1 = 2\pi/\omega_1} + \underbrace{C\sin(\omega_2 t + \phi_2)}_{T_2 = 2\pi/\omega_2} + \underbrace{D\cos\omega_3 t}_{T_3 = 2\pi/\omega_3}$$

$Step\ 1 \rightarrow$ Minimize , T_1, T_2, T_3

$Step\ 2 \rightarrow \frac{T_1}{T_2} \rightarrow Rational$

$\frac{T_1}{T_3} \rightarrow Rational \rightarrow x(t)$ is periodic nature.

$Step\ 3 \rightarrow T_0 = LCM(T_1, T_2, T_2) = \frac{LCM(N^r)}{HCF(D^r)}$

Note:

1. Above procedure is valid only when individual sinusoids are free from external operators.
   $x(t) = |\sin t| + |\cos t| \rightarrow Fails$
2. If above procedure holds true:

$\rightarrow \omega_0 = HCF(w_1, w_2, w_3) = \frac{HCF(N^r)}{LCM(D^r)}$

$\rightarrow \omega_0 = 2\pi / T_0$

$\rightarrow$ $\omega_1 = k_1\omega_0$ $\omega_2 = k_2\omega_0$

k_1^{th} *Harmonic* k_2^{th} *Harmonic*

$\omega_3 = k_3\omega_0$

k_3^{th} *Harmonic*

33. $x(t) = \sin 20t + \cos 50t$ find the fundamental time period of the signal?

Step 1: $T_1 = \frac{2\pi}{20} = \frac{\pi}{10}, T_2 = \frac{2\pi}{50} = \frac{\pi}{25}$

Step 2: $\frac{T_1}{T_2} = \frac{5}{2} \rightarrow R$: x(t) is Periodic

Step 3: $T_0 = LCM\left(\frac{\pi}{10}, \frac{\pi}{25}\right) = \frac{LCM(\pi,\pi)}{HCF(10,25)}$

$= \frac{\pi}{5}$ sec

$\omega_0 = \frac{2\pi}{T_0} = 10$ rad/sec

$\omega_1 = k_1\omega_0$

$20 = k_1(10)$ $\quad k_1 = 2 \rightarrow 2^{nd}$ Harmonic

$\omega_2 = k_2\omega_0$

$50 = k_2 10$ $\quad k_2 = 5 \rightarrow 5^{th}$ Harmonic

Observation:

$t \rightarrow t + \pi/2 = |\cos t| + |\sin t|$

$x(t) = \left|\sin\left(t + \frac{\pi}{2}\right)\right| + \left|\cos\left(t + \frac{\pi}{2}\right)\right|$

$= |\cos t| + |-\sin t|$

$F.T.P. = \pi/2$ sec

Example 1.34:

$x(t) = \max(\sin t, \cos t)$, calculate fundamental time period = ?

Solution:

$$x(t) = \begin{cases} \sin t & : \sin t > \cos t \\ \cos t & : \cos t > \sin t \end{cases}$$

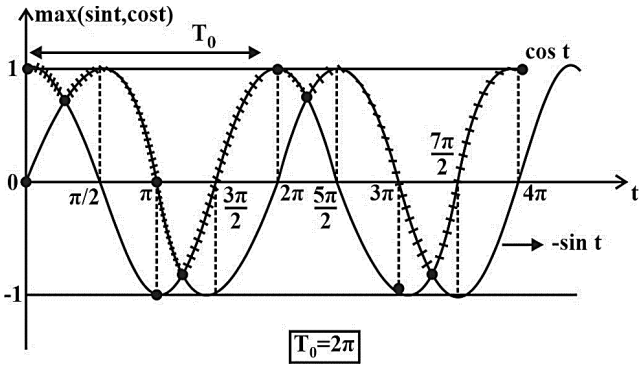


Fig. 1.63

Discrete-Time Periodic Signals

Fundamental time period:

$N_0 \rightarrow$ Smallest assembly duration, which repeats itself.

Or

Minimum no. of samples which repeats itself.

$x(n) = \{\ldots \underset{\uparrow}{1},\ 2,\ 3,\ 1,\ 2,\ 3,\ 1,\ 2,\ 3,\ \ldots\}$

$N_0 = 3$

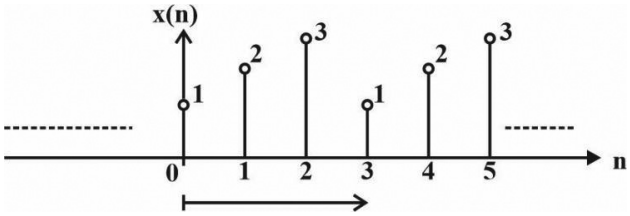


Fig. 1.64

Restrictions on N_0	*Allowed* T_0
1. $N_0 \neq 0$	1. N_0 = constant
2. $N_0 \neq \infty$	2. N_0 = *Integer*
3. $N_0 \neq$ *–ve* 4. N_0 ≠ Rational (p/q) Irrational ($\sqrt{2}, \sqrt{3}, \pi \ldots$)	3. N_0 *may* be function of n

Discrete-time sinusoids and DT complex exponentials:

1. DT sinusoids and DT complex exponentials are not always periodic.

Check: $x(n) = A\cos(\omega_0 n + \phi)$

Step a): $\frac{2\pi}{\omega_0} = N \rightarrow$ *Rational* :

$x(n)$ *is Periodic*

Step b): $N_0 = N \times r$ r: Smallest integer such that N_0 is integer.

Example:

$x(t) = A\cos\left(\frac{5\pi}{7}t\right)$	$x(n) = A\cos\left(\frac{5\pi}{7}n\right)$
1. Periodic 2. $T_0 = \frac{2\pi \times 7}{5\pi} = \frac{14}{5}$	1. Periodic=? 2. N= $\frac{2\pi}{\omega_0} = \frac{14}{5} \rightarrow$ *Rational* $x(n)$ is Periodic 3. $N_0 = N \times r$ $N_0 = \frac{14}{5} \times r = 14$

Note: 'N' strictly must be an integer.

35. Calculate fundamental time period of

$$x(n) = e^{-j\left(\frac{5\pi}{3}n+\frac{\pi}{6}\right)}$$

Solution:

a) Periodic → ?

b) $N = \frac{2\pi}{\omega_0} = \frac{2\pi \times 3}{5\pi} = \frac{6}{5} = Rational \rightarrow$ Periodic

c) $N_0 = N \times r = \frac{6}{5} \times r \rightarrow N_0 = 6$

Periodicity of linear combination of discrete-time signal:

1. Individual DT sinusoids may or may not be periodic.
2. If individual DT sinusoids are periodic, then their combination is also

   Periodic with $N_0 = LCM(N_{01}, N_{02}...)$

x_1	x_2	$x_1 + x_2$
P	P	P
P	NP	NP
NP	P	NP
NP	NP	NP

36. $x(n) = \cos(1.3\pi n) + \cos(1.2\pi nt + 0.3\pi)$
Calculate fundamental time period of the signal x(n)?

Solution:

$N_1 = \frac{2\pi}{1.3\pi} = \frac{20}{13}$ Rational → Periodic

$N_{01} = 20$

$N_2 = \frac{2\pi}{1.2\pi} = \frac{20}{12}$ Rational → Periodic

$N_{02} = 5$

$N_0 = LCM(20, 5) = 20$

Important observation:

$x(t) \rightarrow T_0$ $x(at) \rightarrow \frac{T_0}{\lvert a \rvert}$	$x(n) \rightarrow N_0$ $x(an) \rightarrow ?$
$P \times NP = NP$ $P + NP = NP$ Where NP is not dc	$P \times NP = NP$ $P + NP = NP$ Where NP is not dc

Periodicity of sampled version of CT sinusoid:

$$x(t) \xrightarrow{t = nT_s} x(nT_s)$$

$x(t) = A\cos(\omega_0 t) \rightarrow T_0 = 2\pi / \omega_0$

$x(nT_s) = A\cos(\omega_0 nT_s)$

↓

$x(n) = A\cos(n\omega_0 T_s) \rightarrow N = \frac{2\pi}{\omega_0 T_s} \rightarrow$ Rational

Condition for x(n) to be periodic:

$N = \frac{2\pi}{\omega_0 T_s} = \frac{T_0}{T_s} \rightarrow$ Rational

$\frac{\text{Fundamental Time Period of CTS}}{\text{Sampling Interval}} \rightarrow$ Rational

Note:

$\frac{T_s}{T_0} \rightarrow Rational$ $\frac{f_s}{f_0} \rightarrow Rational$

$\frac{f_0}{f_s} \rightarrow Rational$

Exceptional cases:

$x(t) = \cos\left(\frac{\pi}{8}t^2\right)$	$x(n) = \cos\left(\frac{\pi}{8}n^2\right)$
↓	↓
Non periodic	?

Note:

1. $A\cos(\omega_0 t^k + \phi)$ $\begin{matrix} k = 1 \nearrow & P \\ k \neq 1 \searrow & NP \end{matrix}$

2. $A\cos(\omega_0 n^k + \phi) \xrightarrow{\text{what ever is } k}$ **always verify**

$k = 1 \rightarrow$ Standard Procedure

$k \neq 1 \rightarrow x(n+N) = x(n)$

$N_{min} = ? = N_0$

38. Calculate fundamental time period of $x(n) = \cos\left(\frac{\pi}{8}n^2\right)$?

Solution:

$x(n+N) = x(n)$

$$\cos\left(\frac{\pi}{8}(n+N)^2\right) = \cos\left(\frac{\pi}{8}n^2\right)$$

$$\frac{\pi}{8}(n+N)^2 = \frac{\pi}{8}n^2 + 2\pi k$$

$$\frac{\pi}{8}n^2 + \frac{\pi}{8}N^2 + \frac{\pi}{4}nN = \frac{\pi}{8}n^2 + 2\pi k$$

$$\frac{\pi}{8}N^2 + \frac{\pi}{4}nN = 2\pi k$$

$$\boxed{N^2 + 2nN = 16k} \qquad n, k \in I$$

$N = 1 \quad 1 + 2n = 16k$

$N = 4 \quad 16 + 8n = 16k$

$N = 8 \quad 64 + 16n = 16k \quad N = 8$

Orthogonal, Non-Orthogonal Signals

Case 1:

$x_1(t), x_2(t)$ are CTS

$x_1(t), x_2(t)$ are NP.

$\int_{-\infty}^{\infty} x_1(t)x_2^*(t)dt = 0 \rightarrow x_1(t) and\ x_2(t)$ *are orthogonal.*

above integration can be zero if

a) $x_1(t)x_2^*(t) = 0$

b) $x^*_1(t)x_2(t) = 0$

c) $\int_{-\infty}^{\infty} x_1(t)x_2^*(t)dt = 0$

Case 2:

$x_1(t)$ and $x_2(t)$ are CTS

$x_1(t)$ and $x_2(t)$ are periodic with common fundamental period T_0

$\int_{<T_0>} x_1(t)x_2^*(t)dt = 0 \rightarrow x_1(n) and\ x_2(n) are\ orthogonal.$

Case 3:

$x_1(n)$ and $x_2(n)$ are DTS

$x_1(n), x_2(n)$ are non-periodic.

$\sum_{n=-\infty}^{\infty} x_1(n)x_2^*(n) = 0 \rightarrow$

$x_1(n) and\ x_2(n) are\ orthogonal.$

Case 4:

$x_1(n)$ and $x_2(n)$ are DTS

$x_1(n)$ and $x_2(n)$ are having common time period $= N_0$

$\sum_{<N_0>} = x_1(n)x_2^*(n) = 0 \rightarrow$

$x_1(n) and\ x_2(n) are\ orthogonal.$

Example: $u(t), u(-t)$

1. Non-periodic
2. $\int_{-\infty}^{\infty} u(t)u^*(-t)dt = \int_{-\infty}^{\infty} u(t)u(-t)dt = 0$

   Orthogonal

Example: $u(n), u(-n)$

1. Non-periodic
2. $\sum_{n=-\infty}^{\infty} u(n)u^*(-n) = \sum_{n=-\infty}^{\infty} u(n)u(-n)$

   $= \sum_{n=-\infty}^{\infty} \delta(n) = 1 \neq 0$

   Non-orthogonal

Few important integrations:

1. $\int_{\left\langle\frac{2\pi}{\omega_0}\right\rangle} \sin(\omega_0 t + \phi)dt \quad OR$

   $\int_{\left\langle\frac{2\pi}{\omega_0}\right\rangle} \cos(\omega_0 t + \phi)dt = 0$

2. $\int_{\left\langle\frac{2\pi}{\omega_0}\right\rangle} \sin(k\omega_0 t + \phi)dt \quad OR$

   $\int_{\left\langle\frac{2\pi}{\omega_0}\right\rangle} \cos(k\omega_0 t + \phi)dt = 0$

3. $\int_{\left\langle\frac{2\pi}{\omega_0}\right\rangle} \sin^2(k\omega_0 t + \phi)dt \quad OR$

   $\int_{\left\langle\frac{2\pi}{\omega_0}\right\rangle} \cos^2(k\omega_0 t + \phi)dt = \frac{\pi}{\omega_0}$

Important conclusions:

1. $x_1(t) = A\cos(\omega_1 t + \phi_1)$

$x_2(t) = B\cos(\omega_2 t + \phi_2)$

$\omega_1 = k\omega_0$

$\omega_2 = m\omega_0$

If $\omega_1 \neq \omega_2$

So x_1, x_2 are orthogonal in $\left\langle \frac{2\pi}{\omega_0} \right\rangle$

$\rightarrow$ *irrespective* of ϕ_1, ϕ_2

2. $x_1(t) = A\cos(\omega_1 t + \phi_1)$

$x_2(t) = B\cos(\omega_2 t + \phi_2)$

$\omega_1 = k\omega_0$

$\omega_2 = m\omega_0$

$\rightarrow \omega_1 = \omega_2$

$\rightarrow \Delta\phi = \pm(2n+1)\frac{\pi}{2}$

So x_1 and x_2 are orthogonal in $\left\langle \frac{2\pi}{\omega_0} \right\rangle$.

3. DC signal and $A\cos(\omega_1 + \phi)$ is always orthogonal in $\langle 2\pi / \omega_0 \rangle$ where $\omega_1 = k\omega_0$

Energy, Power, Neither Energy Nor Power [NENP] Signals

1. Neither energy nor power signals

$x(t) / x(n) \rightarrow \pm\infty$ at any signal instant as

$-\infty < t < \infty$

$-\infty < n < \infty$

a) $\delta(t)$ b) $\frac{1}{t}$ c) $t\,\text{ant}$

d) $r(t)$ e) $\ln t$ f) $\frac{1}{n}$

g) e^{t} h) e^{-t}

2. Energy signals: energy signals must have finite energy for infinite possible duration.

$E \rightarrow Joule$

$P \rightarrow \frac{Joule}{\text{sec}}$

$$P = \frac{\overset{\text{Finite}}{\overset{\uparrow}{E}}}{\underset{\infty}{\underset{\downarrow}{T}}} = 0$$

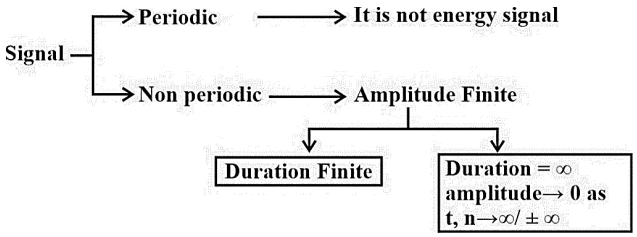


Fig. 1.65

Formula:

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, dt$$

Note: $|\mathbf{x(t)}|^2 = \mathbf{x^2(t)}$ if $\mathbf{x(t)}$ is real.

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \begin{Bmatrix} \ldots |x(-2)|^2 + |x(-1)|^2 \\ + |x(0)|^2 + |x(1)|^2 + \ldots \end{Bmatrix}$$

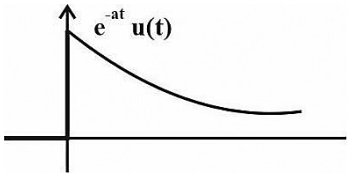
Note: $|\mathbf{x(n)}|^2 = \mathbf{x^2(n)}$ if $\mathbf{x(n)}$ is real.

39. Calculate energy of

$x(t) = (p + jq)e^{-at}u(t) \quad : a > 0$

Solution:

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, dt = \int_{-\infty}^{\infty} \left|(p + jq)e^{-at}u(t)\right|^2 dt$$



$\boxed{|z_1 z_2| = |z_1||z_2|}$

$$= \int_{-\infty}^{\infty} |p + jq|^2 \left|e^{-at}u(t)\right|^2 dt \qquad |p + jq| = \sqrt{p^2 + q^2}$$

$$= \int_{-\infty}^{\infty} (p^2 + q^2)e^{-2at}u^2(t)\,dt \qquad u^2(t) = u(t)$$

$$E_x = \int_{-\infty}^{\infty} (p^2 + q^2)e^{-2at}u(t)\,dt$$

$$E_x = (p^2 + q^2)\int_{-\infty}^{\infty} e^{-2at}dt = (p^2 + q^2)$$

$$\frac{\left(e^{-2at}\right)_0^{\infty}}{-2a} = \frac{(p^2 + q^2)\left(1 - e^{-2a(\infty)}\right)}{2a}$$

$$E_x = \frac{\left(p^2 + q^2\right)}{2a} Joule$$

Note:

$(A+B)^2 = A^2 + B^2 + AB + AB$

$|A+B|^2 = |A|^2 + |B|^2 + AB^* + A^*B$

CT energy sig

$x(t) = x_1(t) + x_2(t)$ Energy of x(t) = ?

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, dt = \int_{-\infty}^{\infty} \left|x_1(t) + x_2(t)\right|^2 dt$$

$$E_x = \int_{-\infty}^{\infty} \left|x_1(t)\right|^2 dt + \int_{-\infty}^{\infty} \left|x_2(t)\right|^2 dt + \int_{-\infty}^{\infty} x_1(t)x_2^*(t)dt + \int_{-\infty}^{\infty} x_1^*(t)x_2(t)dt$$

$$E_x = E_{x_1} + E_{x_2} + \int_{-\infty}^{\infty} x_1 x_2^* dt + \int_{-\infty}^{\infty} x_1^* x_2 dt$$

Note:

1. **If $x_1(t)$ and $x_2(t)$ are orthogonal.**

   $E_x = E_{x_1} + E_{x_2}$

2. **$x(t) = \pm x_1(t) \pm x_2(t) \pm \cdots \pm x_k(t)$**

   Where $x_1(t), x_2(t) \ldots x_k(t)$ are forming set of mutually orthogonal signal.

   $E_x = E_{x_1} + E_{x_2} + \ldots\ldots + E_{x_k}$

40. Calculate energy of $x(t) = e^{-a|t|} : a > 0$.

Solution:

Method 1→ $E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, dt = \int_{-\infty}^{\infty} e^{-2a|t|} dt$

$$= 2\int_0^{\infty} e^{-2at} dt = \frac{1}{a} \text{ Joule}$$

Method 2→ $x(t) = e^{-at}u(t) + e^{at}u(-t)$

$x_1(t)$ and $x_2(t)$ are orthogonal

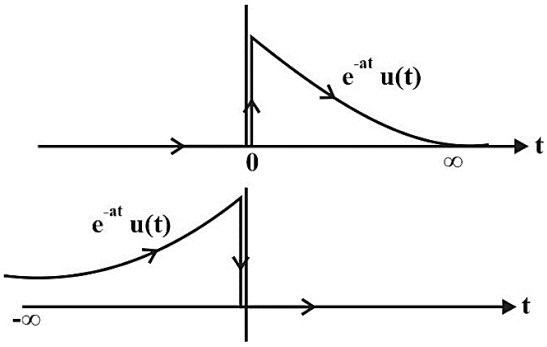
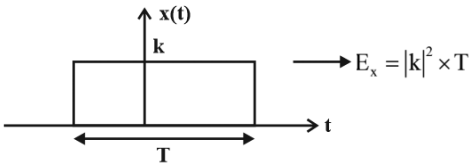
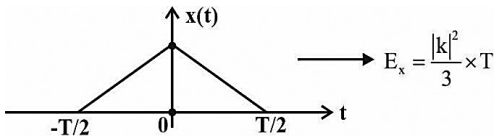


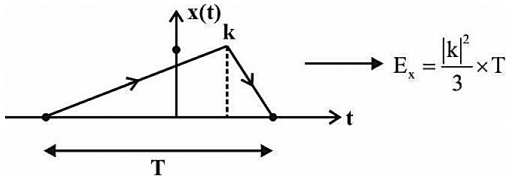
Fig. 1.66

$$E_x = Ex_1 + Ex_2 = \frac{|1|^2}{2a} + \frac{|1|^2}{2a} = \frac{1}{a} J$$

Important results:

1. $x(t) = e^{-at}u(t) : a > 0$
   $\rightarrow E_x = 1/2a$ Joule
2. $x(t) = e^{-at}u(-t) : a > 0$
   $\rightarrow E_x = 1/2a$ Joule
3. $x(t) = e^{-a|t|} : a > 0$
   $\rightarrow E_x = 1/a$ Joule
4. $x(t) = (p + jq)e^{-at}u(t) : a > 0$
   $\rightarrow E_x = p^2 + q^2 / 2a$ Joule
5. $x(t) = (p + jq)e^{at}u(-t) : a > 0$
   $\rightarrow E_x = p^2 + q^2 / 2a$ Joule
6. $x(t) = (p + jq)e^{-a|t|} : a > 0$
   $\rightarrow E_x = (p^2 + q^2) / a$ Joule
7. 

8. 

9. 

10.

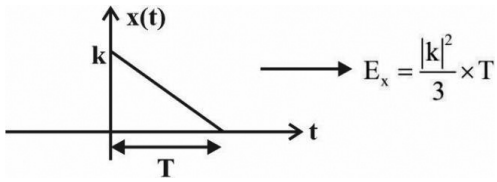
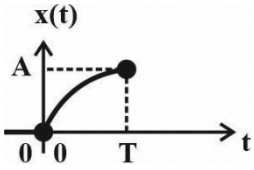


Fig. 1.67

Energy of sinusoidal pulse:

$$x(t) = \begin{cases} A\sin\omega_0 t & : 0 \le t \le T \\ 0 & : \text{elsewhere} \end{cases}$$



$$E_x = \int_{-\infty}^{\infty} x^2(t)dt = \int_0^T A^2 \sin^2 \omega_0 t dt$$

$$= \int_0^T \frac{A^2}{2} dt - \int_0^T \frac{A^2}{2} \cos 2\omega_0 t dt$$

$$Ex = \frac{A^2 T}{2} - \frac{A^2}{4\omega_0} \sin 2\omega_0 T$$

$$2\omega_0 T = m\pi \qquad \omega_0 = 2\pi f_0$$

$$f_0 = \frac{m\pi}{2\pi \times T \times 2} = \qquad \sin\theta = 0 \rightarrow \theta = m\pi$$

$$f_0 = \frac{m}{4T} \quad m \in I$$

$$E_x = \frac{A^2}{2} \times T = \frac{A}{2} \times duration$$

Important observation:

Signal		*Energy*
$x(t)$	$\rightarrow$	E_x
$x(t-t_0)$	$\rightarrow$	E_x
$x(-t)$	$\rightarrow$	E_x
$x(at)$	$\rightarrow$	$\frac{E_x}{\lvert a\rvert}$
$x(-at+b)$	$\rightarrow$	$\frac{E_x}{\lvert a\rvert}$
$-kx(-at+b)$	$\rightarrow$	$\lvert -k\rvert^2 \frac{E_x}{\lvert a\rvert}$

41. Calculate energy of x(t).

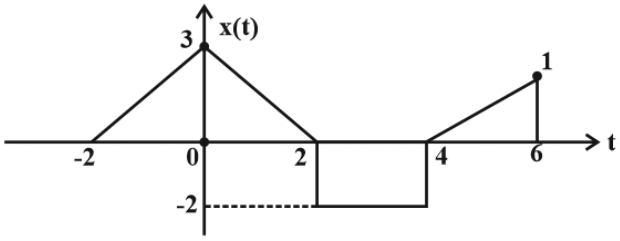


Fig. 1.68

Solution:

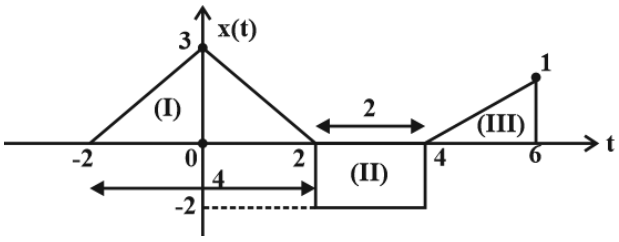


Fig. 1.69

$$E_x = \frac{3^2}{3} \times 4 + (-2)^2 \times 2 + \frac{(1)^2}{3} \times 2 = \frac{62}{3} Joule \quad ...(i$$

Discrete-Time Energy Signals

1. $E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \sum_{n=-\infty}^{-1} |x(n)|^2 + \sum_{n=0}^{\infty} |x(n)|^2$

2. $x(n) = x_1(n) + x_2(n)$

$$E_x = E_{x_1} + E_{x_2} + \sum_{n=-\infty}^{\infty} x_1(n) x_2^*(n)$$

$$+ \sum_{n=-\infty}^{\infty} x_1^*(n) x_2(n)$$

If x_1 and x_2 are orthogonal then

$$\boxed{E_v = E_v + E_v}$$

42. Calculate energy of $x(n) = \left(\frac{1}{2}\right)^{|n|}$.

Solution:

Method 1:

$$E_x = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^{2|n|} = \sum_{n=-\infty}^{\infty} \left(\frac{1}{4}\right)^{|n|}$$

$$= \left(\frac{1}{4}\right)^0 + 2\sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n = 1 + 2\left[(1/4) + (1/4)^2 + \cdots\right]$$

$$= 1 + \frac{2 \times (1/4)}{1 - (1/4)} = 1 + \frac{2}{3} = \frac{5}{3} \text{ Joules}$$

Method 2:

$$x(n)=\left(\frac{1}{2}\right)^{|n|}=\left(\frac{1}{2}\right)^{n}u(n)+\left(\frac{1}{2}\right)^{n}u(-n-1)$$

$$=\begin{cases}(1/2)^{n} & n\geq 0\rightarrow u(n)\\ \left(\frac{1}{2}\right)^{-n} & n<0; n\leq -1,\rightarrow u(-n-1)\end{cases}$$

$$\boxed{E_x=E_1+E_2}$$

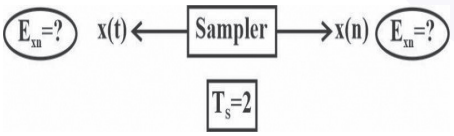
$$=\frac{4}{3}+\frac{1}{3}=\frac{5}{3}J$$

$$E_1=\sum_{n=0}^{\infty}\left(\frac{1}{4}\right)^{n}=4/3$$

$$E_2=\sum_{n=-\infty}^{-1}(2/4)^{-n}=\frac{1/4}{1-1/4}=\frac{1}{3}$$

Energy after sampling:

43. $x(t)=\begin{cases}1-\frac{|t|}{2}: & |t|\leq 2\\ 0 & \text{else}\end{cases}$



Calculate energy of x(n) = ?

Solution:

$$x(n)=\begin{cases}1-\frac{|nT_s|}{2}: & |nT_s|\leq 2\\ 0: & else\end{cases}$$

$$x(n)=\begin{cases}1-|n| & : |n|\leq 1: -1\leq n\leq 1: n=-1,0,1\\ 0 & else.\end{cases}$$

$$x(n)=\{0,\underset{\uparrow}{1},0\}$$

$E_x=1$ Joule

Power signal:

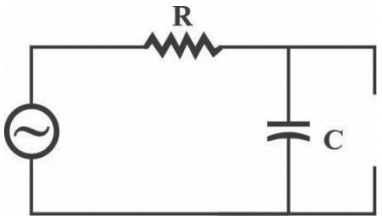




Fig. 1.70

1. Electrical signal or supply
2. Power in supply ≡ power of electrical signal ≡ power in waveforms.

Example: Periodic → NENP signal

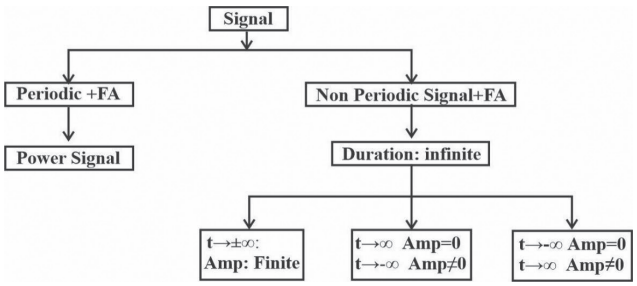


Fig. 1.71

Formulae:

Case 1: x(t) is non-periodic in nature.

$$P_x=\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}|x(t)|^2\,dt=\overline{|x(t)|^2}=MSV[|x(t)|]$$

a) T: symmetrical duration.

b) As long as x(t) is real: $|x(t)|^2=x^2(t)$

44.

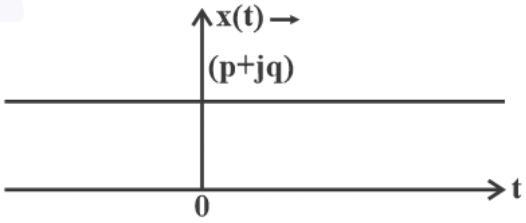


Fig. 1.72

Solution:

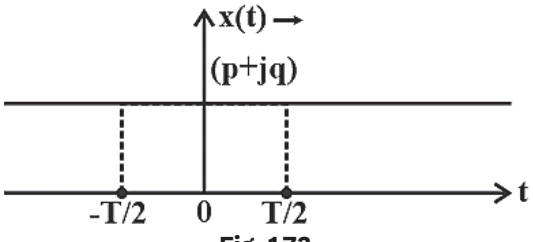


Fig. 1.73

$$P_x=\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}|x(t)|^2\,dt$$

$$= \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}\left(p^2+q^2\right)dt \rightarrow$$

$$= \lim_{T\to\infty}\frac{\left(p^2+q^2\right)}{T}\int_{-T/2}^{T/2}dt = \lim_{T\to\infty}\left(p^2+q^2\right)\frac{T}{T}$$

$$= \left(p^2+q^2\right) \text{ watt}$$

Important observations:

$$\left|Ae^{j\omega t}\right| = |A||e^{j\omega t}| = |A|$$

$$e^{j\omega t} = \cos\omega t + j\sin\omega t$$

$$\left|e^{j\omega t}\right| = \sqrt{\cos^2\omega t + \sin^2\omega t} = 1$$

1. $x(t) = A \rightarrow P_x = |A|^2$
2. $x(t) = (p + jq) \rightarrow p_x = \left(p^2+q^2\right)$
3. $x(t) = Ae^{j\omega t} \rightarrow P_x = |A|^2$
4. $\underset{\text{finite}}{P} = \frac{E \rightarrow \infty}{T \rightarrow \infty} \rightarrow$ Energy of power signals

   $\rightarrow \infty \rightarrow$ arrow symbol align under P.
5. $x(t) \rightarrow P_x$

   $x(-at + b) \rightarrow P_x$

   $-kx(-at + b) \rightarrow |-k|^2 P_x$

Case 2:

x(t) is periodic with T_0

$$P_x = \frac{1}{T_0}\int_{<T_0>}|x(t)|^2\,dt = \frac{1}{T_0}\int_0^{T_0}|x(t)|^2\,dt$$

$$= \frac{1}{T_0}\int_{-T_0/2}^{T_0/2}|x(t)|^2\,dt$$

$$P_x = \frac{1}{T_0}\int_0^{T_0}\left|x\left(t\right)\right|^2 dt = \frac{E_{T_0}}{T_0}$$

E_{T0} = Energy of one time period.

Power of periodic signals:

$$P_x = \frac{1}{T_0}\int_{T_0}|x(t)|^2\,dt = \frac{E_{T_0}}{T_0}$$

$$MSV[|x(t)|] = \frac{E_{T_0}}{T_0} = \overline{|x(t)|^2}$$

$$\sqrt{MSV[|x(t)|]} = \sqrt{\frac{E_{T_0}}{T_0}} = \sqrt{\overline{|x(t)|^2}} = \text{ Root mean}$$

square value of $\left|x\left(t\right)\right|$.

Step 1: Identify one time period $\left(T_0\right)$ of the signal.

Step 2: Calculate energy of *one* $T_0 \rightarrow E_{T_0}$

Step 3: $P_x = \frac{E_{T_0}}{T_0}$

45. Calculate the power of x(t).

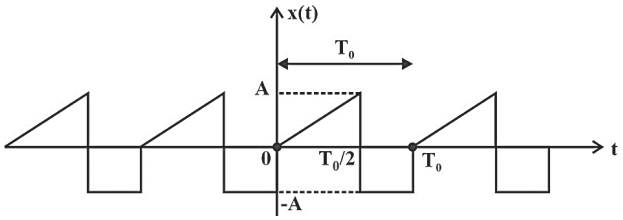


Fig. 1.74

Solution:

S.1 →The signal is periodic with time period T_0.

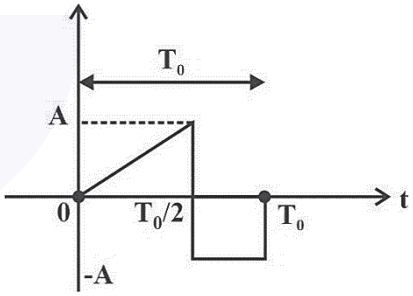


Fig. 1.75

S.2 → $E_{T_0} = \frac{A^2}{3}\times\frac{T_0}{2} + A^2\times\frac{T_0}{2} = \frac{2A^2}{3}T_0$

S.3 → $P_x = \frac{E_{T_0}}{T_0} = \frac{2A^2}{3}$

X(t): Real

1. $P_x = \overline{x^2(t)} = \frac{2A^2}{3} = MSV \text{ of } x(t)$
2. RMS value of $x\left(t\right) = \sqrt{P_x} = A\sqrt{\frac{2}{3}}$
3. Power of $x\left(\frac{1-6t}{4}\right) = P_x = 2A^2/3$
4. RMS value of $x\left(\frac{1-6t}{4}\right) = \sqrt{P_x} = A\sqrt{\frac{2}{3}}$

Note:

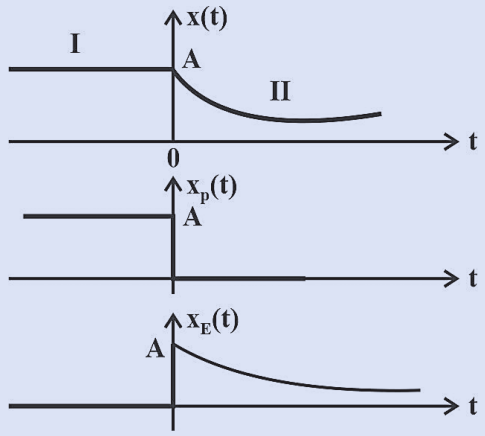


Fig. 1.75

$x(t) = x_E(t) + x_p(t) \rightarrow$ Power signal

$$P_x = P_{x_E} + P_{x_p} + \lim_{\mapsto\to\infty} \frac{1}{T}\int_{-T/2}^{T/2}\left[x_E x_P^{*} + x_E^{*} x_p\right]dt$$

If

1. x_E and x_P are non-overlapping
2. x_E and x_P are orthogonal

$$P_x = P_{x_E} + P_{x_p} = P_{x_p}$$

46. RMS value of x(t) = ?

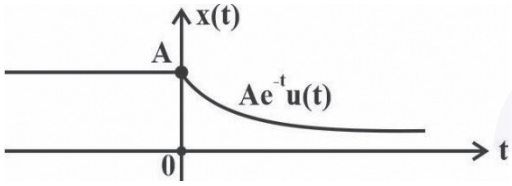


Fig. 1.76

Solution:

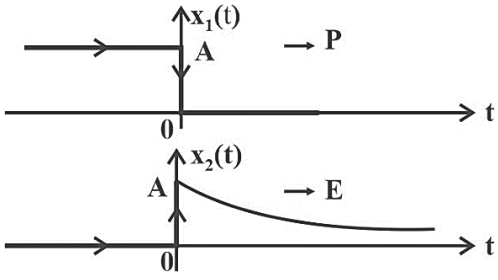


Fig. 1.77

$x(t) = x_E(t) + x_p(t)$

$P_x = 0 + P_{x_p}$

$$P_x = \frac{A^2}{2}$$

$$RMS = \frac{A}{\sqrt{2}}$$

Power of sinusoidal signal:

$x(t) = A\cos\left(\omega_0 t + \phi\right)$

OR

$A\sin\left(\omega_0 t + \phi\right)$

$$P_x = \frac{A^2}{2}$$

$RMS = \sqrt{P_x} = A/\sqrt{2}$

Case 1: $x(t) = A\cos\left(\omega_1 t + \phi_1\right) + B\cos\left(\omega_2 t + \phi_2\right)$

$\omega_1 = m\omega_0 \quad , \omega_2 = k\omega_0$

a) $\omega_1 \neq \omega_2$

b) Irrespective of ϕ_1, ϕ_2

→ $x_1(t)$ *and* $x_2(t)$ are orthogonal

$P_x = P_{x_1} + P_{x_2}$

$$P_x = \frac{A^2}{2} + \frac{B^2}{2}$$

Case 2: $x(t) = A\cos\left(\omega_1 t + \phi_1\right) + B\cos\left(\omega_2 t + \phi_2\right)$

$\omega_1 = m\omega_0 \ \omega_2 = k\omega_0$

a) $\omega_1 = \omega_2$

b) $\Delta\phi \rightarrow \left(2n+1\right)\frac{\pi}{2}$

→ $x_1(t)$ *and* $x_2(t)$ are orthogonal

$$P_x = P_{x_1} + P_{x_2} = \frac{A^2}{2} + \frac{B^2}{2}$$

Case 3:

$x(t) = A\cos\left(\omega_1 t + \phi_1\right) + B\cos\left(\omega_2 t + \phi_2\right)$

$x_1(t), x_2(t)$: Non-orthogonal $\omega_1 = k\omega_0$

$\omega_2 = m\omega_0$

$$P_x = \frac{A^2}{2} + \frac{B^2}{2} + AB\cos\left(\phi_2 - \phi_2\right)$$

Case 4:

$x(t) = r_0 + r_1\cos\left(\omega_0 t + \phi_1\right) + r_2\cos\left(2\omega_0 t + \phi_2\right)$

$+ r_3\cos\left(3\omega_0 t + \phi_3\right) + ...$

$$P_x = r_0^2 + \frac{r_2^2}{2} + \frac{r_2^2}{2} + \cdots\cdots$$

Discrete-time power signal:

Case 1:

x(n) is non-periodic

$$P_x = \lim_{N\to\infty} \frac{1}{(2N+1)} \sum_{n=-N}^{N} |x(n)|^2$$

N: Symmetric duration

Case 2:

x(n) is non-periodic with N_0

$$P_x = \frac{1}{N_0} \sum_{n=0}^{N_0-1} |x(n)|^2 = \frac{E_{N_0}}{N_0}$$

N_0: Fundamental time period.

47. Calculate the power of $x(n) = A(1)^n = A$.

Solution: Note: $\sum_{a}^{b} 1 = (b-a)+1$

$$P_x = \lim_{N\to\infty} \frac{1}{(2N+1)} \sum_{n=-N}^{N} |A|^2$$

$$P_x = \lim_{N\to\infty} \frac{1}{(2N+1)} |A|^2 \sum_{n=-N}^{N} 1$$

$$= \lim_{N\to\infty} \frac{1}{(2N+1)} |A|^2 \times (2N+1) = |A|^2$$

Note:

$x(t) \to E$	$x(n) \to E$
$x(-at+b) \to \frac{E}{\lvert a\rvert}$	$x(-at+b) \to$ **cant say**
$x(t) \to P$	$x(n) \to P$
$x(-at+b) \to P$	$x(-an+b) \to$ **cant say**

Causal, non-causal, anti-causal signal:

Causal signal:

$x(t) = 0 \quad for \ t < 0$

$x(n) = 0 \quad for \ n < 0;\ n \le -1$

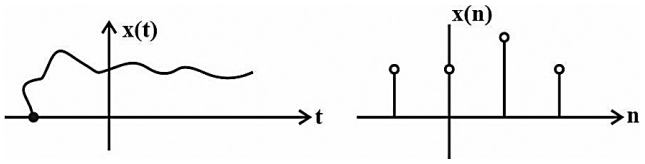
Example:

$u(t)$

$u(t-2)$

$u(n-2)$

Non-causal signal: Which is not causal.



Anti-causal signal:

$x(t) = 0 \quad for \quad t > 0$

$x(n) = 0 \quad for \quad n \ge 0$

Example:

$u(t) \to$ Causal signal

$u(-r) \to$ Anti-causal signal

$u(n) \to$ causal signal

$u(-n) \to$ non-causal signal

$u(-n-1) \to$ Anti-causal signal

Integrable vs absolutely integrable:

$x(t) \to \int_{-\infty}^{\infty} x(t)dt \to finite \to x(t)$ is Integrable

$x(t) \to \int_{-\infty}^{\infty} |x(t)|dt \to finite \to x(t)$ is absolutely Integrable

Summable vs absolutely summable:

$x(n) \to \sum_{n=-\infty}^{\infty} x(n) \to x(n)$: *summable*

$x(n) \to \sum_{n=-\infty}^{\infty} |x(n)| \to x(t)$ is absolutely *summable*

Bounded and Unbounded Signal

Bounded and unbounded signals do not have any correlation with absolutely integrable or absolutely summable nature of signal.

Bounded signal: (Amplitude bounded signal)

$|x(t)| \le M < \infty$ if $-\infty < t < \infty$

$|x(n)| \le M < \infty$ if $-\infty < n < \infty$

Unbounded signal: Which is not bounded.

Bounded signal	Unbounded signal
$\cos t$ / $\sin t$	$\delta(t)$
$Sgn(t)$	$r(t)$
$u(t)$	Int
dc	e^{at}
$e^{-a\|t\|}$ $a>0$	Int
$\delta(n)$	$\tan t$

Note:

Finite → Finite duration

Bounded → Bounded in amplitude

Practice Questions

Problems:

1. The function shown in the figure can be represented as:

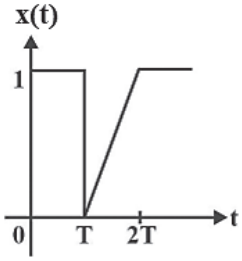


Fig. 1.78

(a) $u(t)-u(t-T)+\dfrac{(t-T)}{T}u(t-T)$
$-\dfrac{(t-2T)}{T}u(t-2T)$

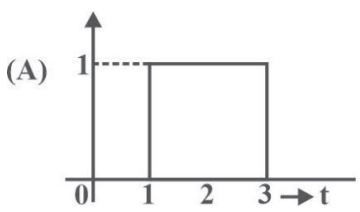
(b) $u(t)+\dfrac{t}{T}u(t-T)-\dfrac{t}{u}u(t-2T)$

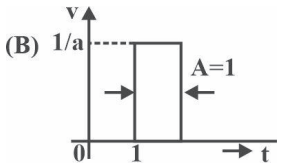
(c) $u(t)-u(t-T)+\dfrac{(t-T)}{T}u(t)$
$-\dfrac{(t-2T)}{T}u(t)$

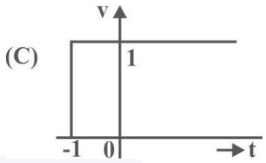
(d) $u(t)+\dfrac{(t-T)}{T}u(t-T)$
$-2\dfrac{(t-2T)}{T}u(t-2T)$

2. Match list 1 with list 2 and select the correct answer using the codes given below the lists:

List 1







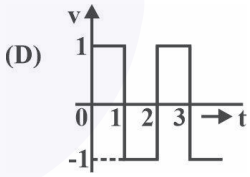


Fig. 1.79

List 2

1. $v(t)=u(t+1)$
2. $v(t)=u(t)-2u(t-1)$
   $+2u(t-2)-2u(t-3)$
3. $v(t)=u(t-1)-u(t-3)$
4. $\underset{a\to 0}{Lt}\, v(t)=\delta(t-1)$

Codes:

	A	B	C	D
(A)	1	2	3	4
(B)	3	4	1	2
(C)	4	3	2	1
(D)	4	3	1	2

3. If a plot of signal x(t) is as shown in the fig. 1,

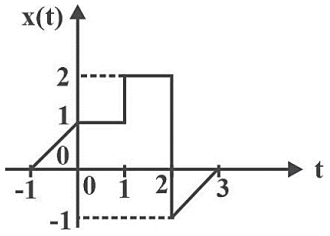
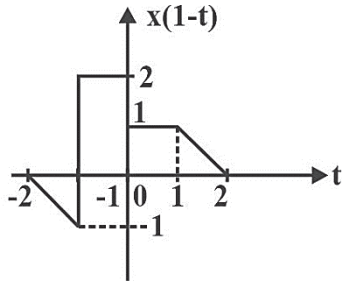


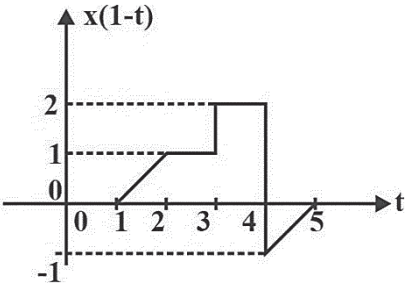
Fig. 1.80

Then the plot of the signal x(1-t) will be

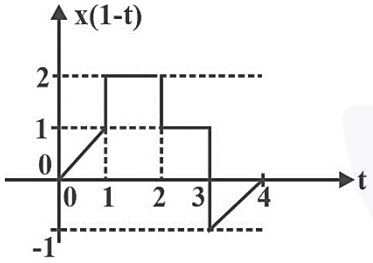
(A)



(B)



(C)



(D)

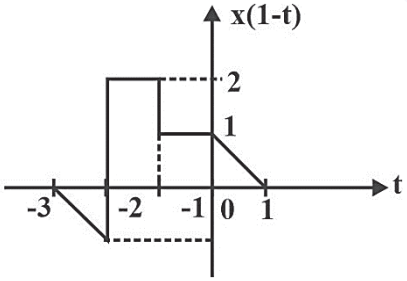


Fig. 1.81

4. In the graph shown in fig., which one of the following expresses v(t)?

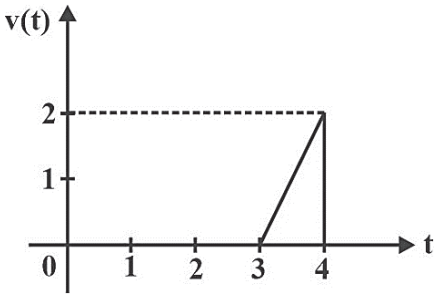


Fig. 1.82

(A) $(2t+6)[u(t-3)+2u(t-4)]$

(B) $(-2t-6)[u(t-3)+u(t-4)]$

(C) $(-2t+6)[u(t-3)+u(t-4)]$

(D) $(2t-6)[u(t-3)-u(t-4)]$

5. The expression for the waveform in terms of step function is given by:

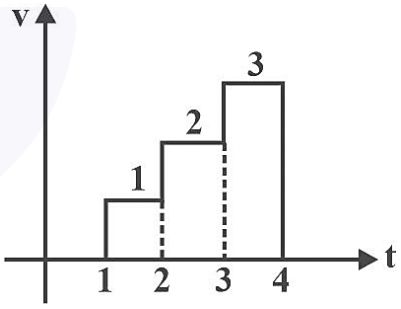


Fig. 1.83

(A) $v = u(t-1) - u(t-2) + u(t-3)$

(B) $v = u(t-1) + u(t-2) + u(t-3)$

(C) $v = u(t-1) + u(t-2) - u(t-3)$

(D) $v = u(t-1) + u(t-2) + u(t-3) - 3u(t-4)$

Key and Explanations

1. (A)

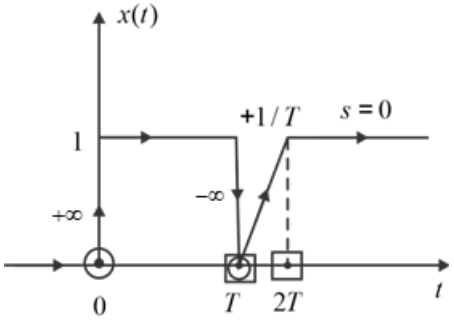


Fig. 1.84

$$x(t) = u(t) - u(t-T) + \frac{1}{T}r(t-T) - \frac{1}{T}r(t-2T)$$

$$= u(t) - u(t-T) + \frac{1}{T}(t-T)\cdot u(t-T) - \frac{1}{T}(t-2T)\cdot u(t-2t)$$

Hence, correct option is (A).

2. (B)

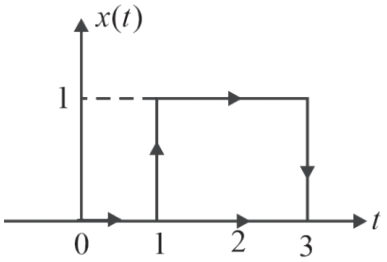


Fig. 1.85

$x(t) = u(t-1) - u(t-3)$

$\rightarrow B.P \ at \ t = 1, 3.$

Hence, (B) is correct.

(B)

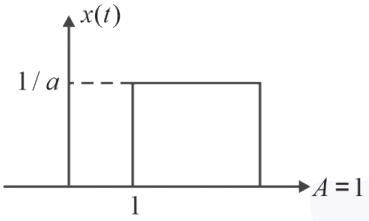


Fig. 1.86

By basic definition of impulse,

Hence (4) is correct code.

$V(t) = \delta(t-1)$ code =

(C)

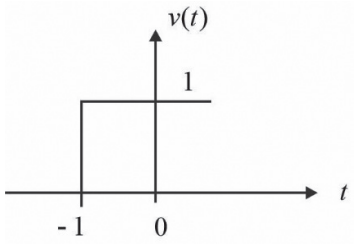


Fig. 1.87

$V(t) = u(t+1)$

By time advancing $u(t)$ by '1'

Hence, (1) is correct code.

(D)

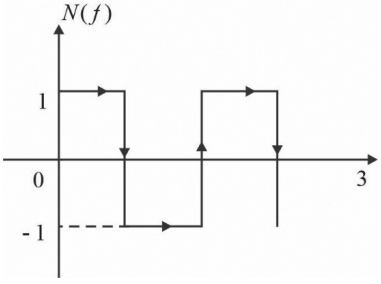


Fig. 1.88

$u(t) = u(t) - 2u(t-1) + 2u(t-2) - u(t-3)$

Hence, (2) is correct code.

3. (A)

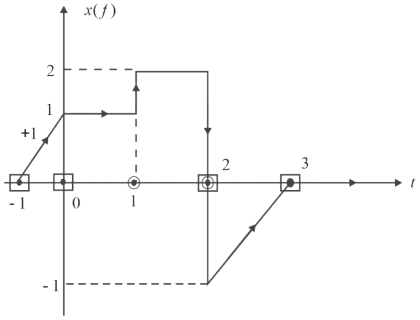


Fig. 1.89

$\boxed{\bullet} \rightarrow$ slop change at BP

$\odot \rightarrow$ height change at *BP*



$x(t+1)$

time advancing $x(t)$ by '1'

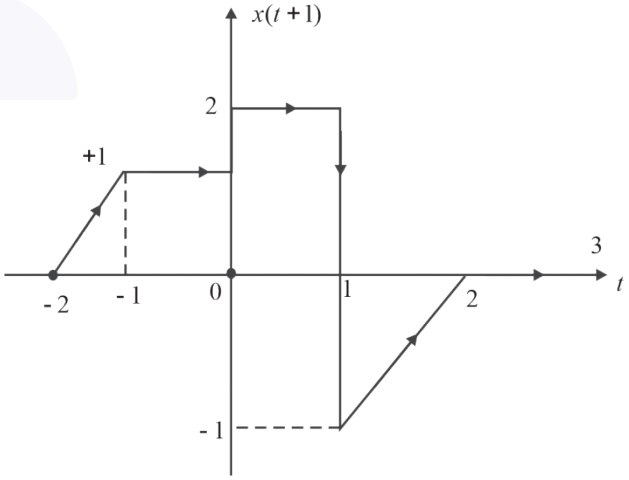


Fig. 1.90

$\underset{x(-t+1)}{\downarrow}$ time reversal

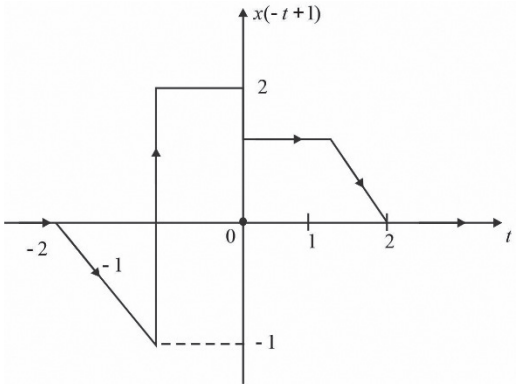


Fig. 1.91

Option (A) is correct.

4. (D)

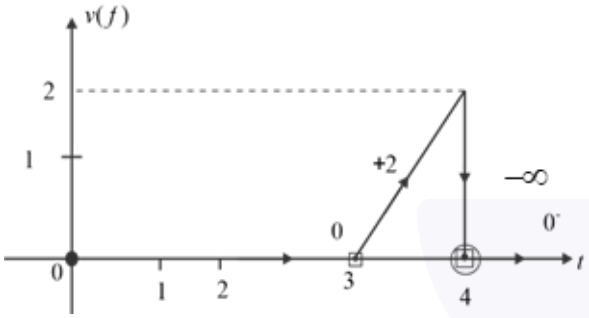


Fig. 1.92

$\boxdot$ ⟶ slop change at *B.P*

$\odot$ ⟶ hight change at *B.P*

$u(t) = 2r(t-3) - 2r(t-4) - 2u(t-4)$

$= 2(t-3)u(t-3) - 2(t-4)u(t-4) - 2u(t-4)$

$x(t) = 2(t-3)u(t-3) - (2(t-4)+2)u(t-1)$

$x(t) = (2t-6)u(t-3) - (2t-6)u(t+1)$

$2(t-6)\big(u(t-3) - u(t-4)\big)$

Option (D) is correct.

5. (D)

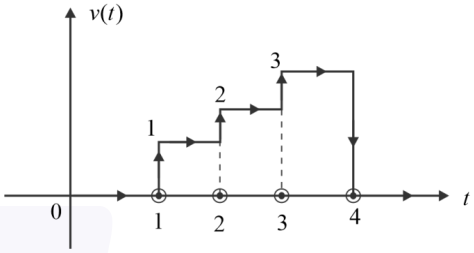


Fig. 1.93

$\odot$ ⟶ slope change at *B.P*.

$u(t) = u(t-1) + u(t-2) + u(t-3) - 3u(t-4)$

Option (D) is correct.